

# **CSE 311:**

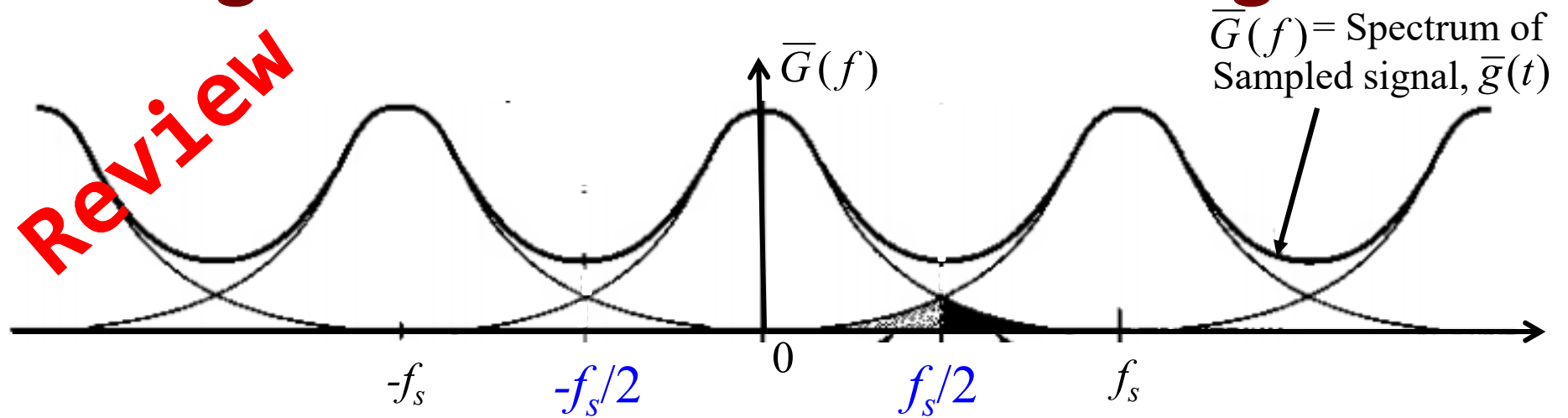
# **Data Communication**

**Instructor:**

**Dr. Md. Monirul Islam**

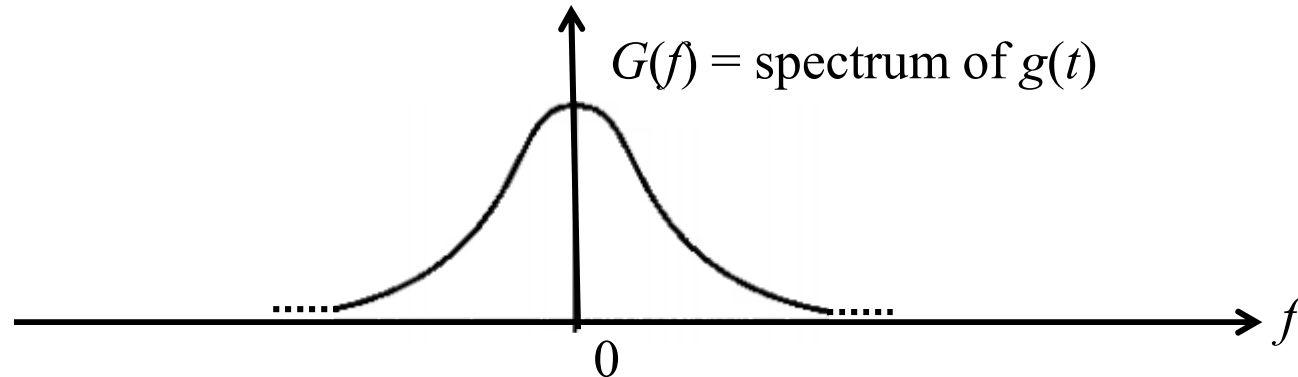
# **Sampling and Analog-to-Digital Conversion**

# Sampling Effect: Non-band-limited signal to band-limited signal



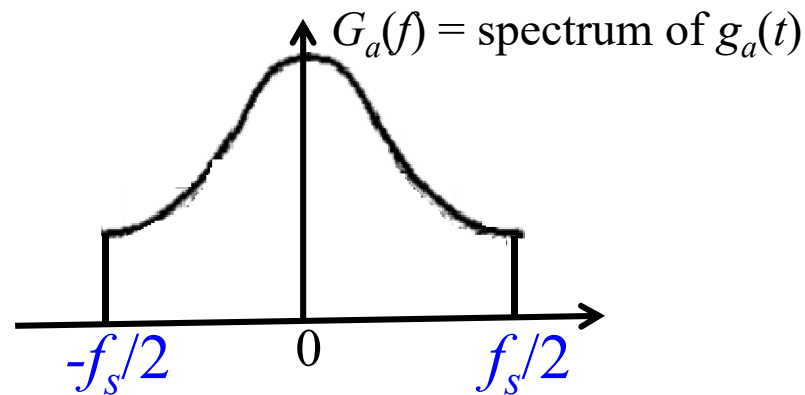
Sampling a non-band-limited signal  $g(t)$  at  $f_s$  is equivalent to Nyquist sampling of some signal  $g_a(t)$  band-limited to  $f_s/2$

# Sampling Effect: Non-band-limited signal to band-limited signal



Let  
sub-Nyquist Sampling of  $g(t)$  at  $f_s$  generates samples  $g(0), g(T_s), g(2T_s), g(3T_s), \dots$

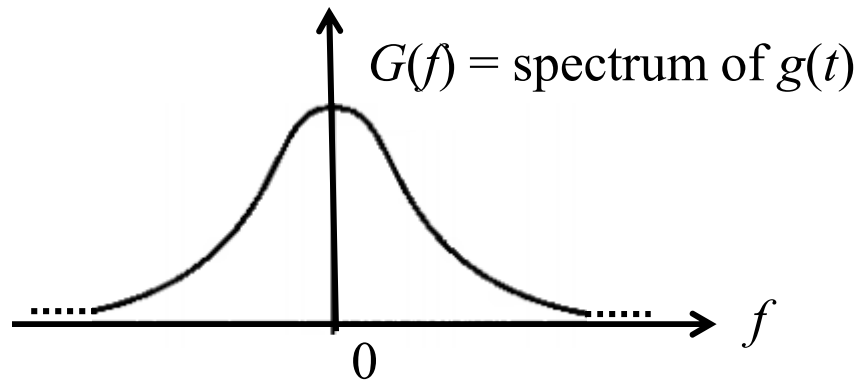
# Sampling Effect: Non-band-limited signal to band-limited signal



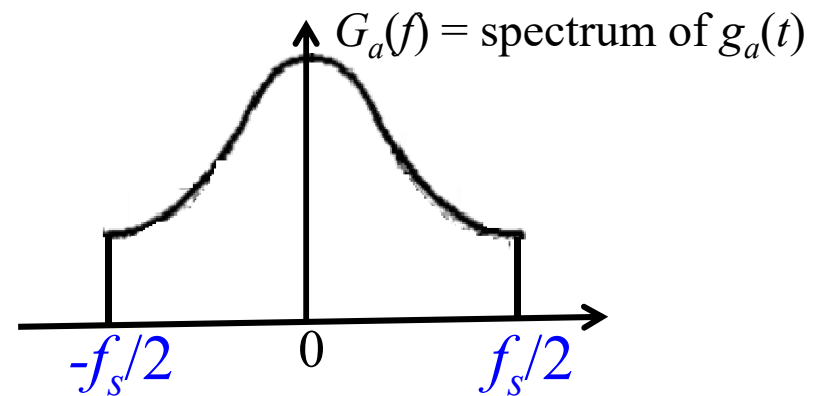
Let

Nyquist Sampling of  $g_a(t)$  at  $f_s$  generates samples  $g_a(0), g_a(T_s), g_a(2T_s), g_a(3T_s), \dots$

# Sampling Effect: Non-band-limited signal to band-limited signal

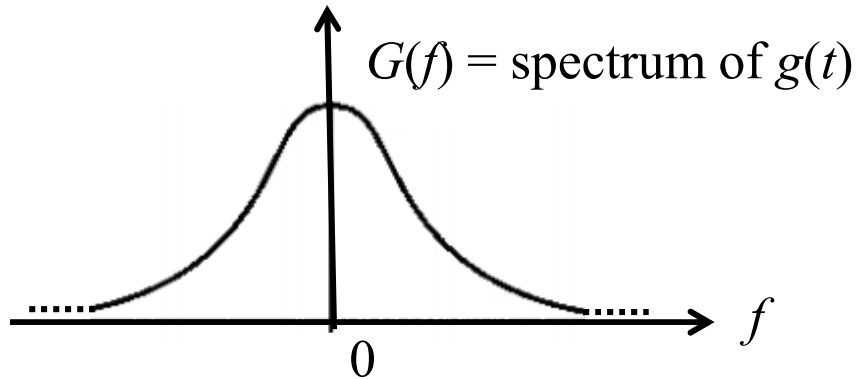


sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$

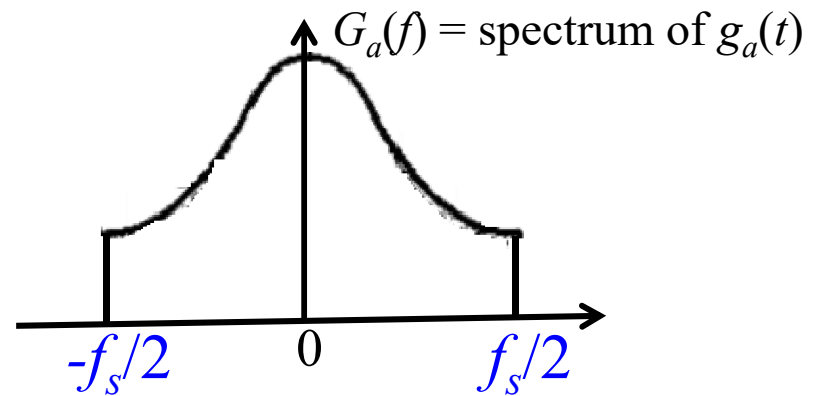


Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$

# Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$

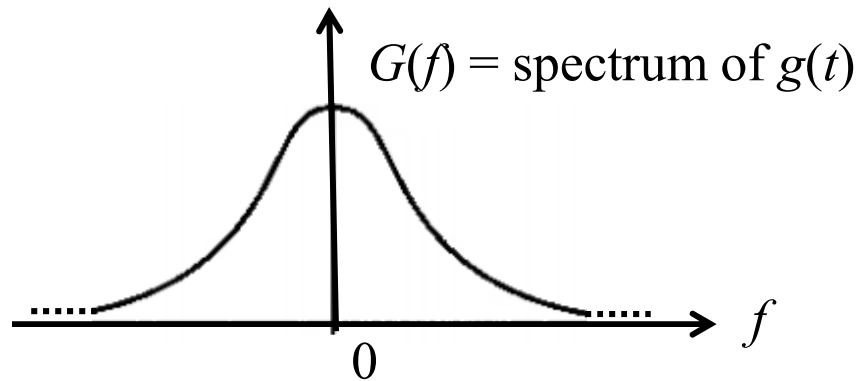


Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$

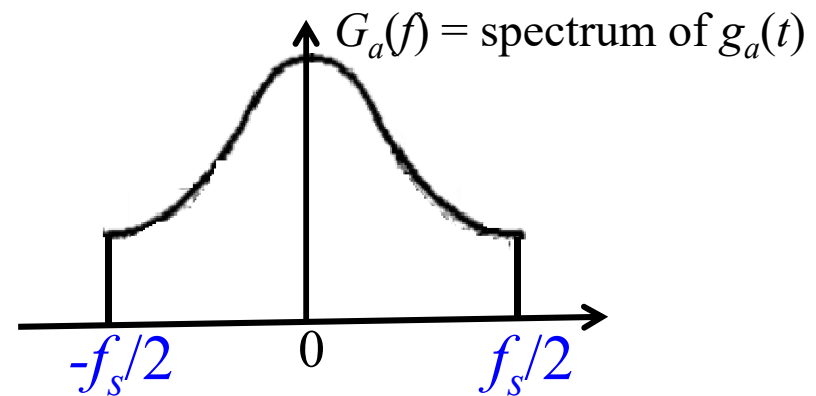
According to sampling effect that we saw,

$$g(0) = g_a(0), g(T_s) = g_a(T_s), g(2T_s) = g_a(2T_s), g(3T_s) = g_a(3T_s), \text{ and so on}$$

# Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$



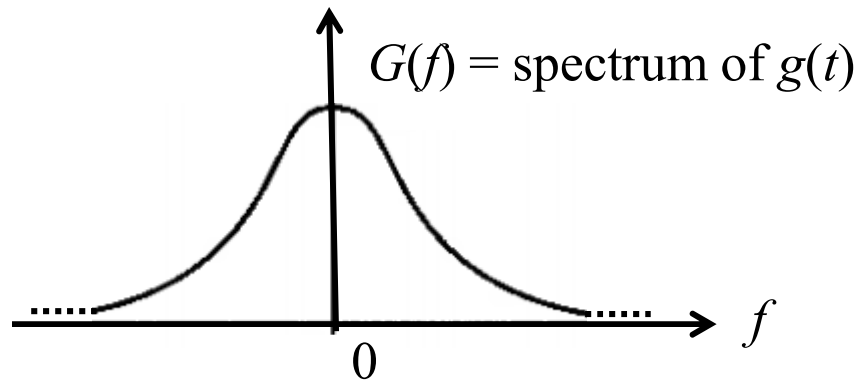
Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$

Therefore,

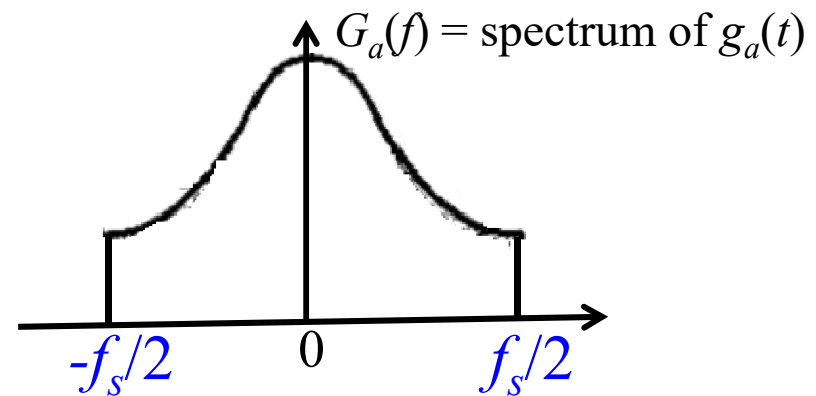
$$g(nT_s) = g_a(nT_s) = g_n$$



# Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$



Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$

Therefore,

$$g(nT_s) = g_a(nT_s) = g_n$$

In other words, sampling  $g(t)$  and  $g_a(t)$  at the rate of  $f_s = 1/T_s$  will generate the **same independent** data sequence

# Maximum Information Rate of a Channel with BW = $B$ Hz

Assume

- Error free, noise less channel
- Channel bandwidth is  $B$

# Maximum Information Rate of a Channel with BW = $B$ Hz

Assume

- Error free, noise less channel
- Channel bandwidth is  $B$

We will prove

- Maximum  $2B$  pieces of information can be sent per second

# Maximum Information Rate of a Channel with BW = $B$ Hz

Assume

- Error free, noise less channel
- Channel bandwidth is  $B$

Information rate

- Channel can send a low pass signal of  $B$  Hz
- This signal can be recovered from samples uniformly taken at  $2B$  samples per second

# Maximum Information Rate of a Channel with BW = $B$ Hz

Assume

- Error free, noise less channel
- Channel bandwidth is  $B$

Information rate

- Channel can send a low pass signal of  $B$  Hz
- This signal can be recovered from samples uniformly taken at  $2B$  samples per second
- This means,  $2B$  samples/second can be sent through the channel

# Maximum Information Rate of a Channel with BW = $B$ Hz

Assume

- Error free, noise less channel
- Channel bandwidth is  $B$

Information rate

- Channel can send a low pass signal of  $B$  Hz
- This signal can be recovered from samples uniformly taken at  $2B$  samples per second
- This means,  $2B$  samples/second can be sent through the channel

We have to prove that

- A sequence of data samples at the rate of  $2B$  samples/second can come from uniform sampling of a signal of bandwidth  $B$  Hz
- The signal can be recovered from these samples

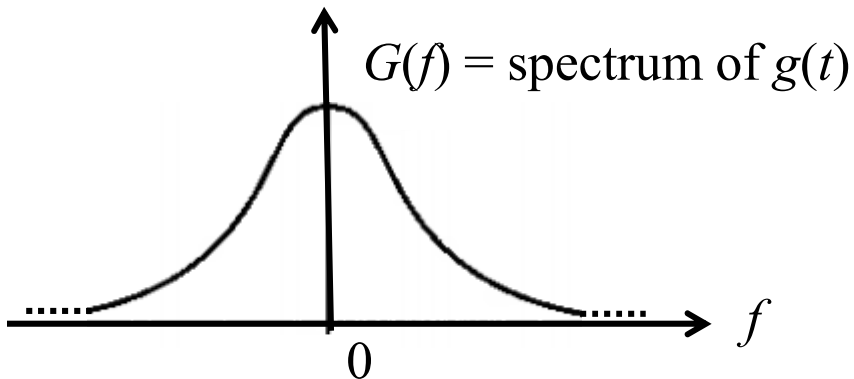
# Maximum Information Rate of a Channel with BW = $B$ Hz

Assume a sequence of samples  $g_0, g_1, g_2, g_3, \dots$  denoted as  $\{g_n\}$  at the rate of  $2B$  s/s

# Maximum Information Rate of a Channel with BW = $B$ Hz

Assume a sequence of samples  $g_0, g_1, g_2, g_3, \dots$  denoted as  $\{g_n\}$  at the rate of  $2B$  s/s

We will always find a signal  $g(t)$  whose samples  $g(0), g(T_s), g(2T_s), g(3T_s), \dots$  matches with  $\{g_n\}$ .

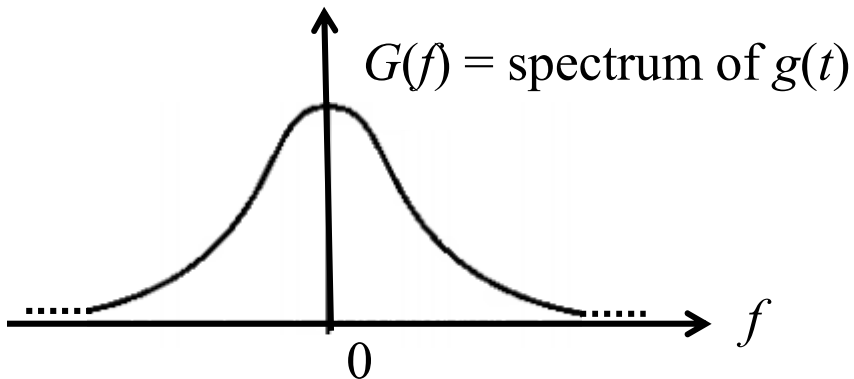




# Maximum Information Rate of a Channel with BW = $B$ Hz

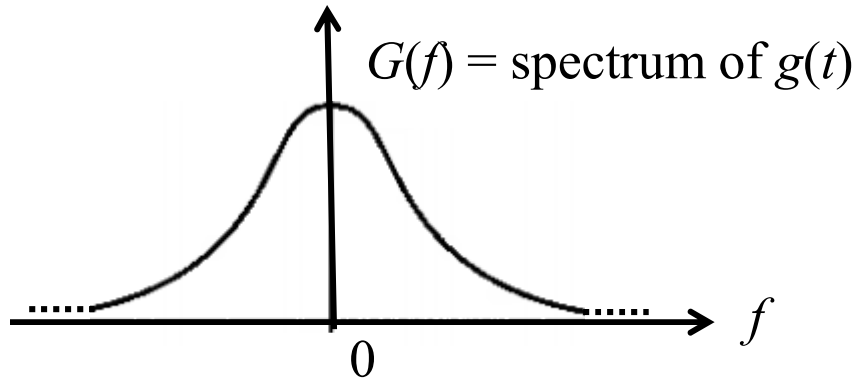
Assume a sequence of samples  $g_0, g_1, g_2, g_3, \dots$  denoted as  $\{g_n\}$  at the rate of  $2B$  s/s

We will always find a signal  $g(t)$  whose samples  $g(0), g(T_s), g(2T_s), g(3T_s), \dots$  matches with  $\{g_n\}$ .

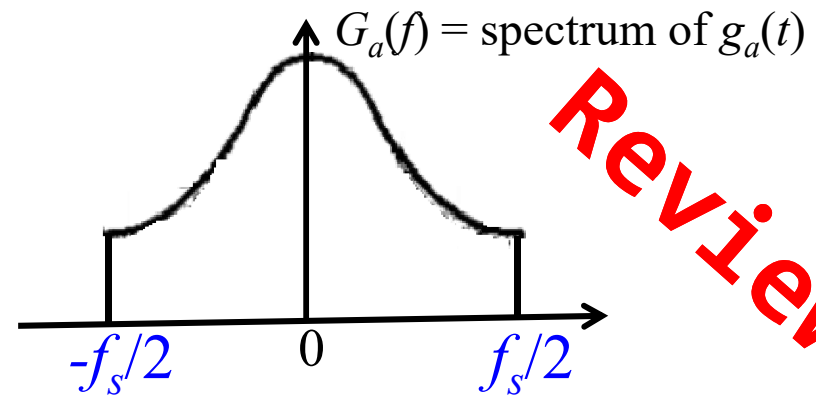


This means,  
 $g_n = g(nT_s)$

# Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$



**Review**

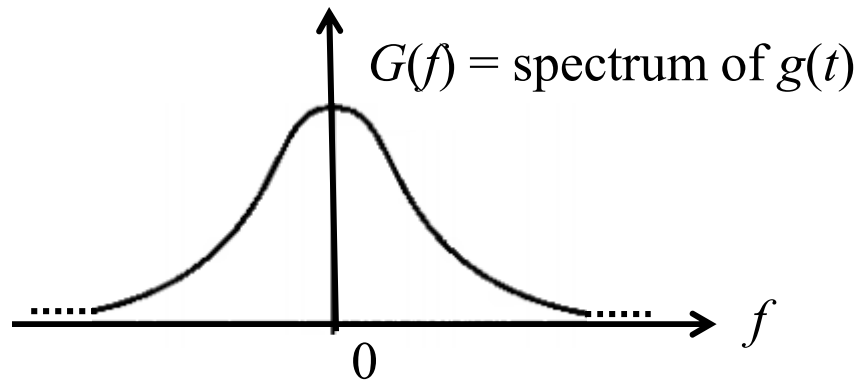
Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$

Therefore,

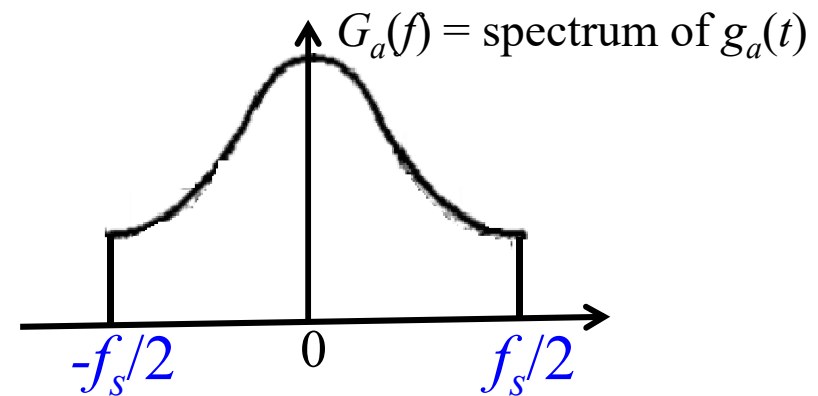
$$g(nT_s) = g_a(nT_s) = g_n$$

In other words, sampling  $g(t)$  and  $g_a(t)$  at the rate of  $f_s = 1/T_s$  will generate the same independent data sequence

# Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$

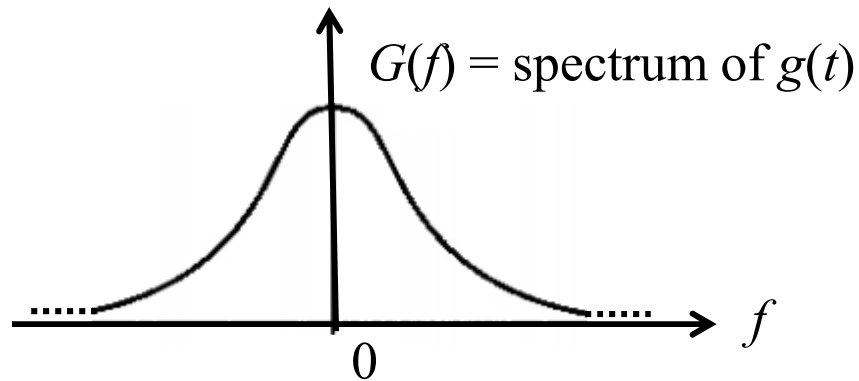


Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$

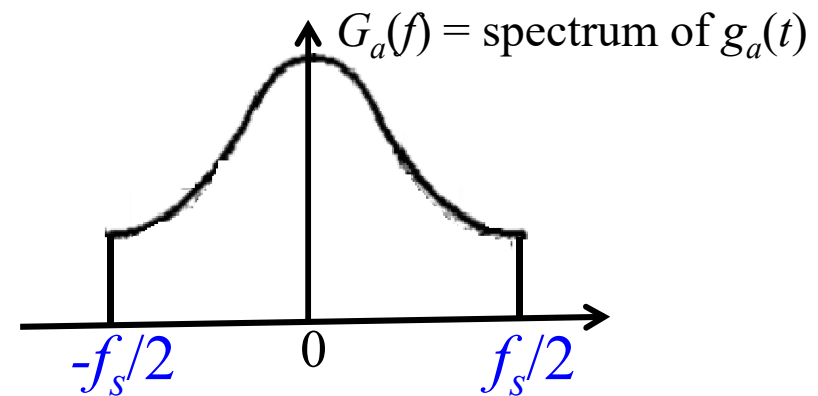


$$\bar{g}(t) = \sum_n g(nT_s) \delta(t - nT_s)$$

# Sampling Effect: Non-band-limited signal to band-limited signal

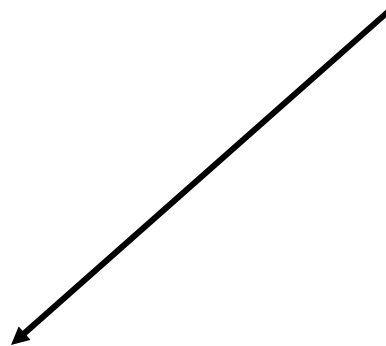


sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$

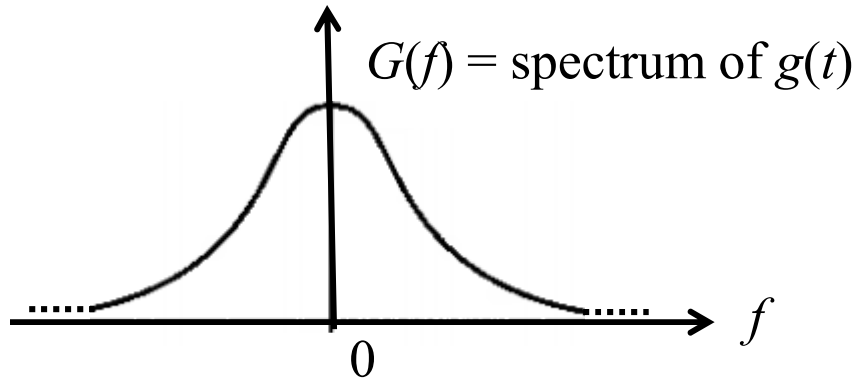


Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$

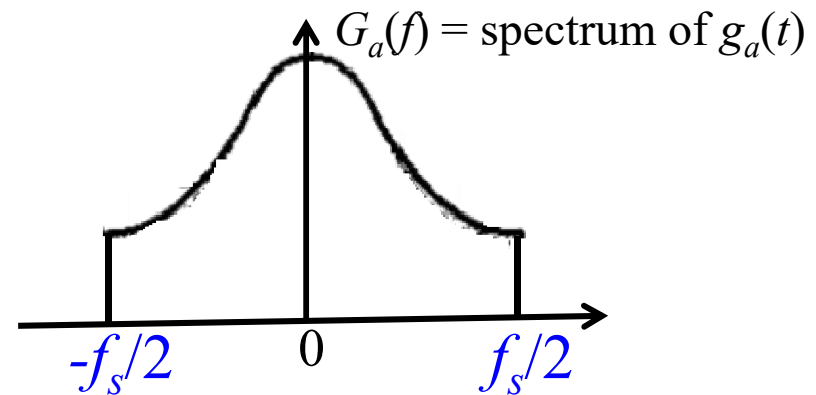
$$\begin{aligned}\bar{g}(t) &= \sum_n g(nT_s) \delta(t - nT_s) \\ &= \sum_n g_a(nT_s) \delta(t - nT_s)\end{aligned}$$



# Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ ,  $\dots$

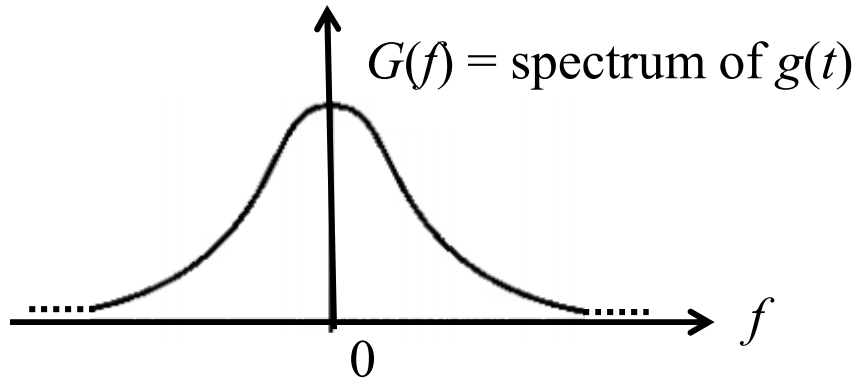


Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ ,  $\dots$



$$\begin{aligned}\bar{g}(t) &= \sum_n g(nT_s) \delta(t - nT_s) \\ &= \sum_n g_a(nT_s) \delta(t - nT_s) \\ &= \sum_n g_n \delta(t - nT_s)\end{aligned}$$

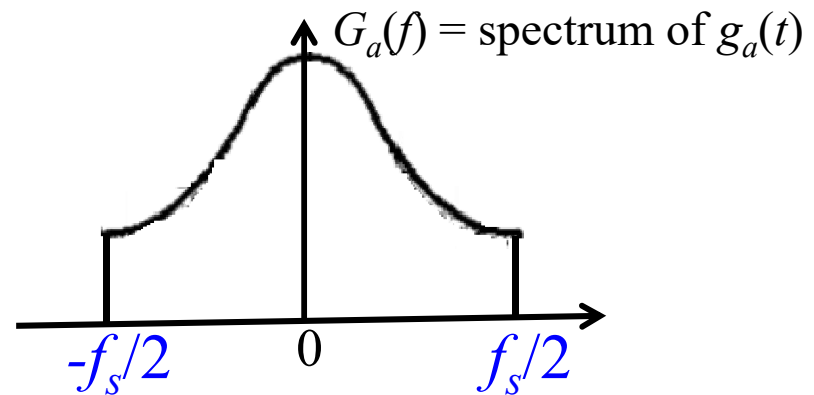
# Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples  $g(0)$ ,  
 $g(T_s)$ ,  $g(2T_s)$ ,  $g(3T_s)$ , ...



$$\begin{aligned}\bar{g}(t) &= \sum_n g(nT_s) \delta(t - nT_s) \\ &= \sum_n g_a(nT_s) \delta(t - nT_s) \\ &= \sum_n g_n \delta(t - nT_s)\end{aligned}$$

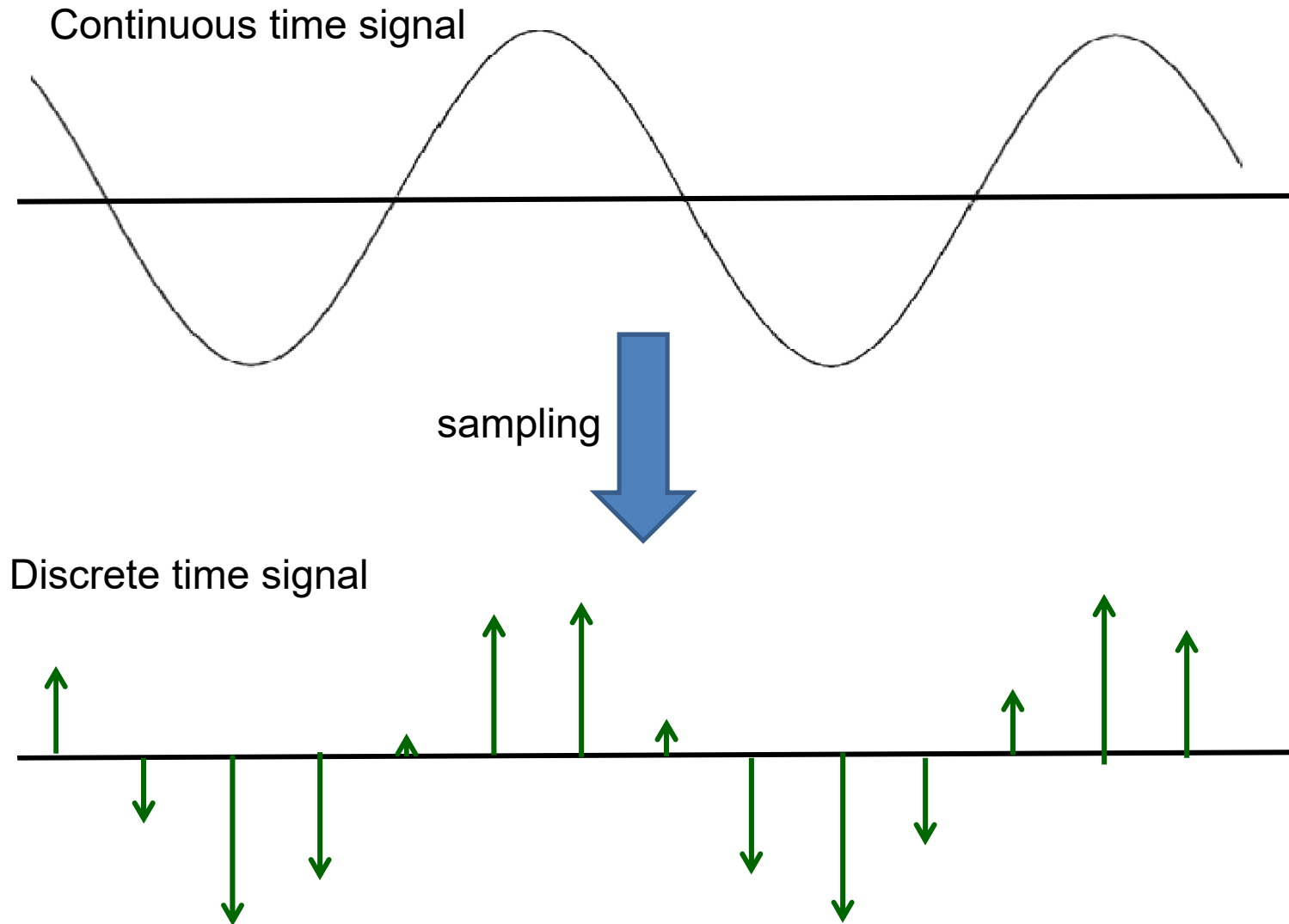


Nyquist Samples  $g_a(0)$ ,  $g_a(T_s)$ ,  $g_a(2T_s)$ ,  
 $g_a(3T_s)$ , ...

To recover  $g_a(t)$ , we can use  $\{g_n\}$  using,

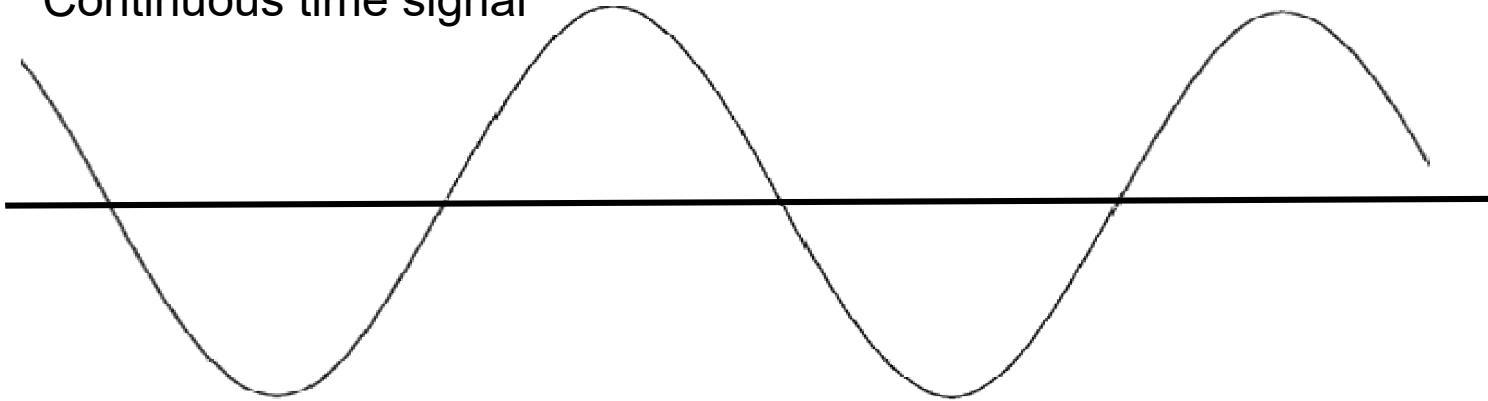
$$g_a(t) = \sum_n g_n \text{sinc}(2\pi Bt - n\pi)$$

# Application of Sampling Theorem

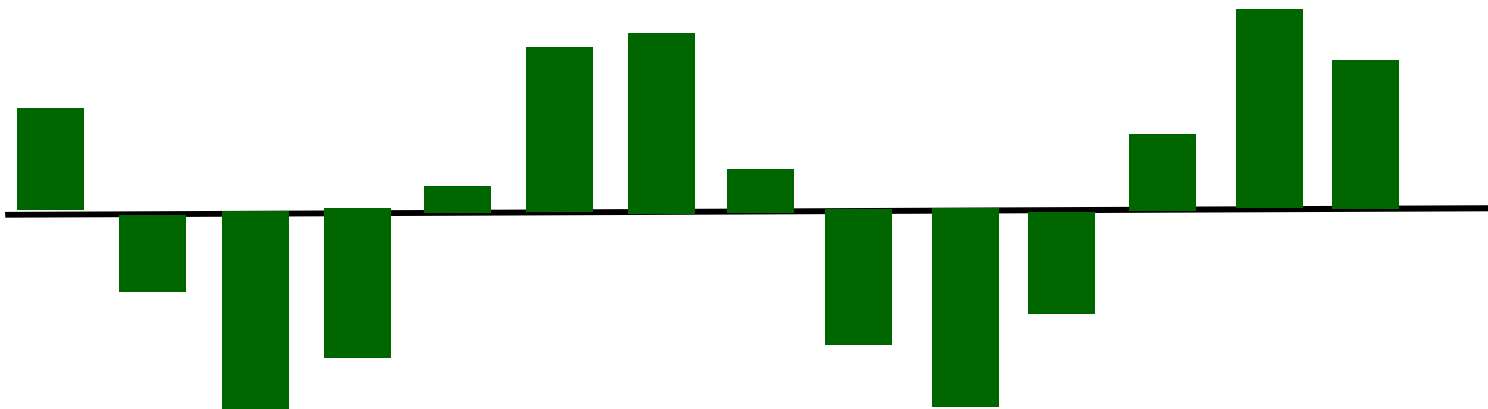


# Application of Sampling Theorem

Continuous time signal

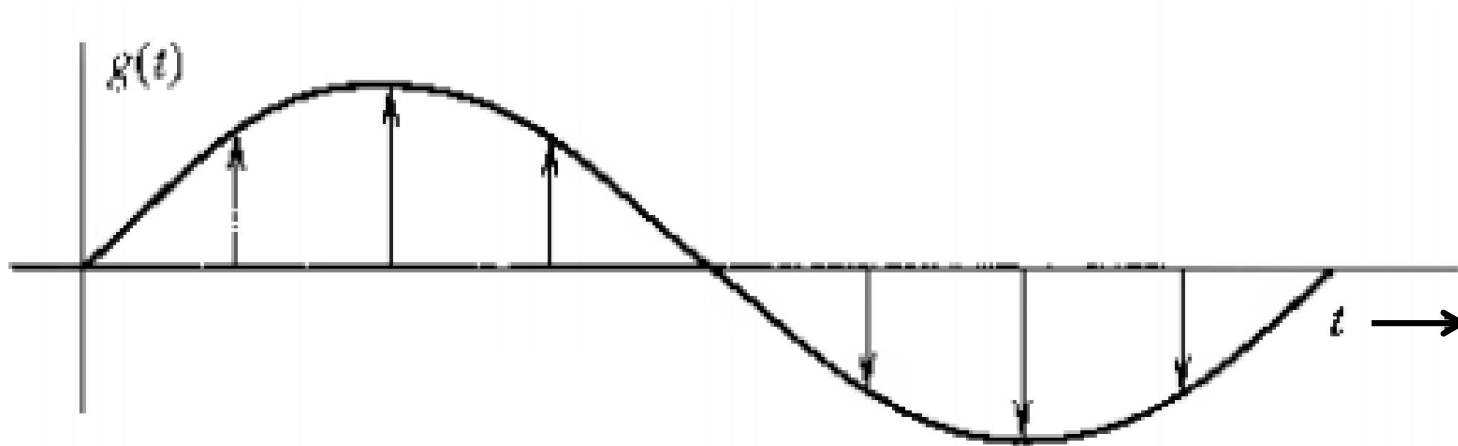


can be represented by pulse train and transmitted thereafter



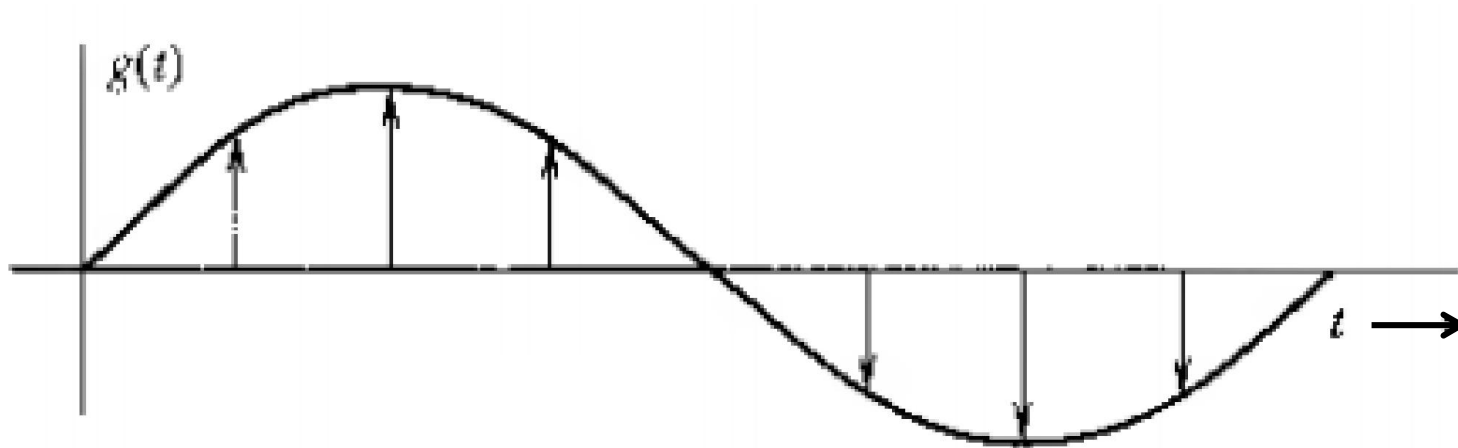


# Application of Sampling Theorem

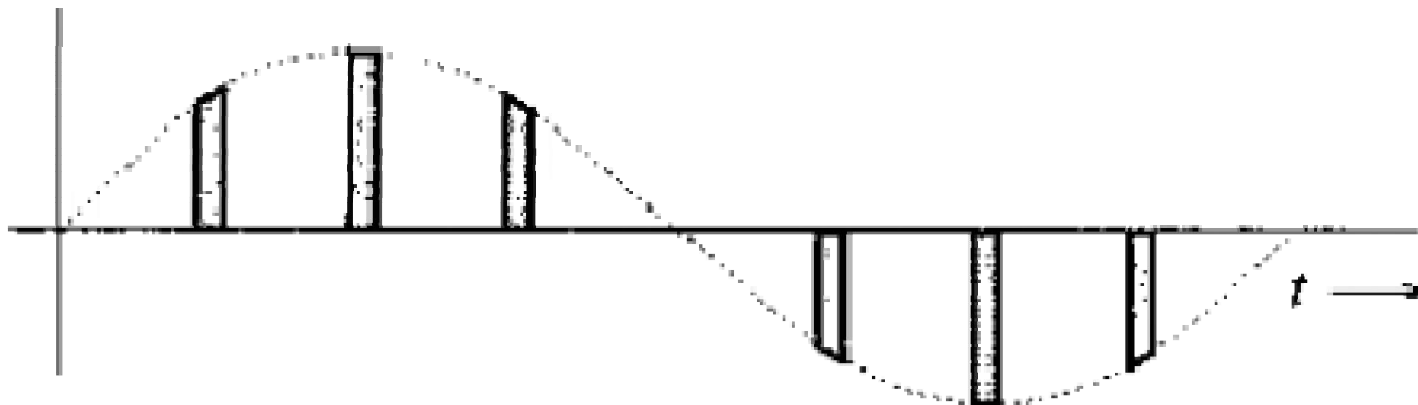


**Pulse modulations:** different ways to transmit sampled signal  
modifying pulse trains

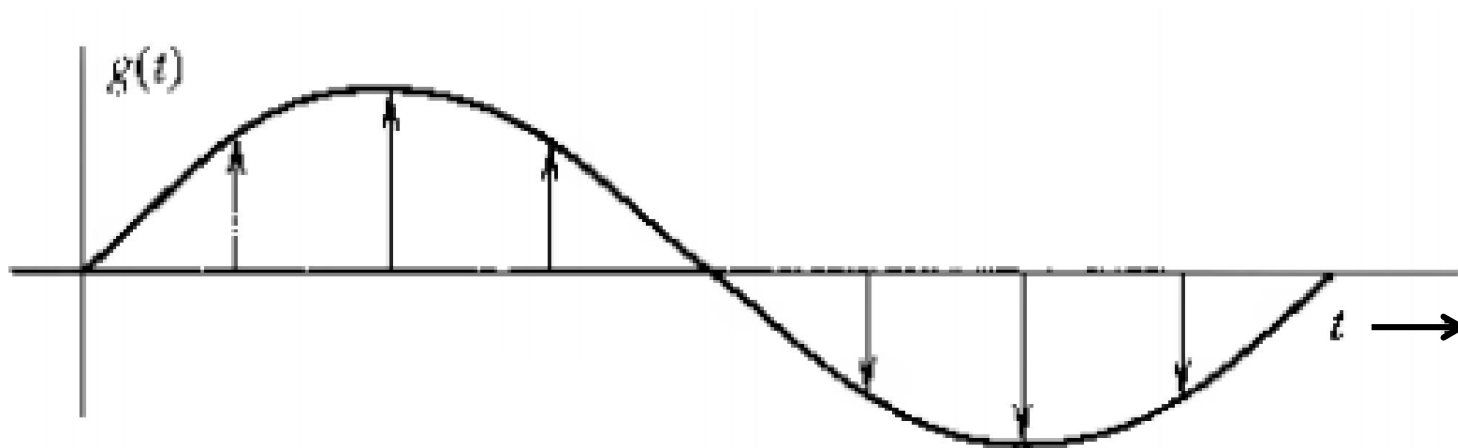
# Application of Sampling Theorem



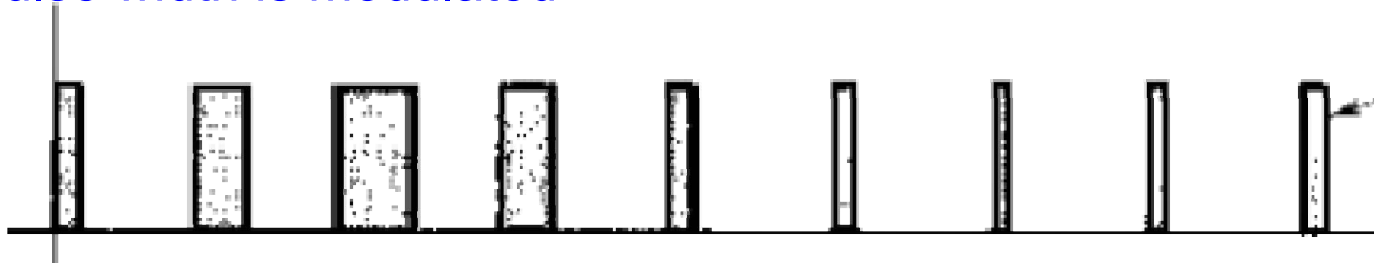
PAM: Pulse amplitude is modulated



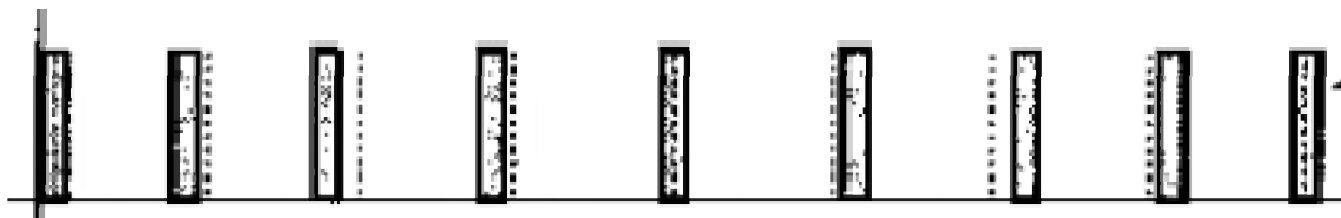
# Application of Sampling Theorem



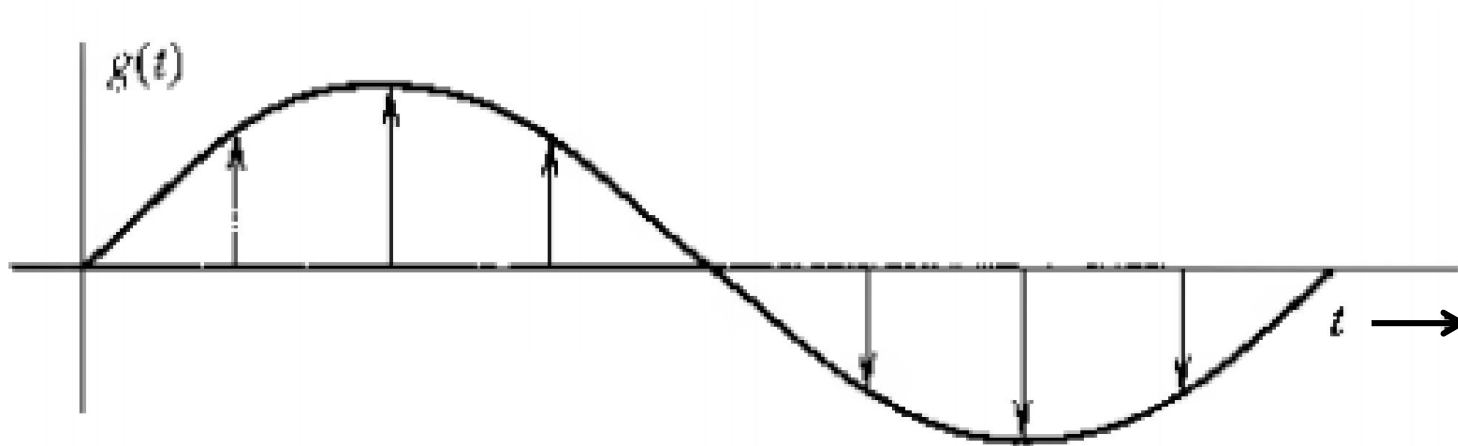
PWM: Pulse width is modulated



PPM: Pulse position is modulated



# Application of Sampling Theorem

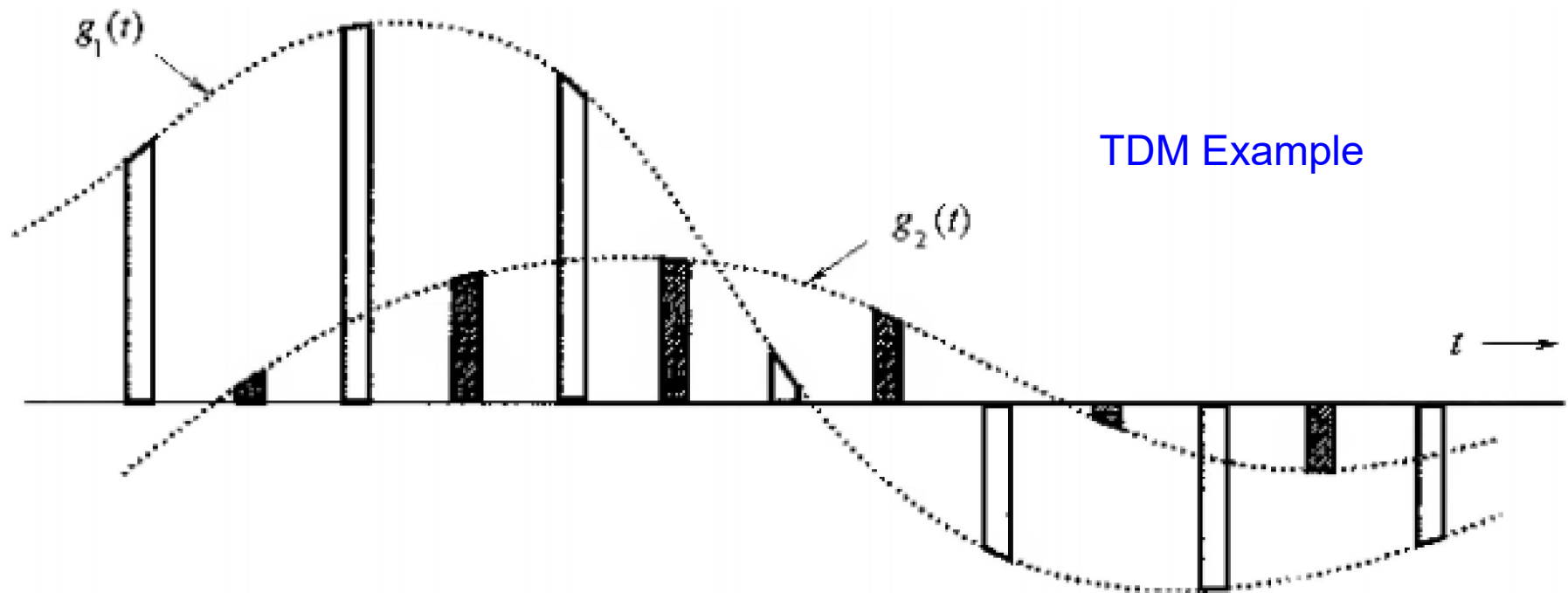


## Pulse code modulation:

- most widely used pulse modulation
- each sample value is converted to a set of pulses.

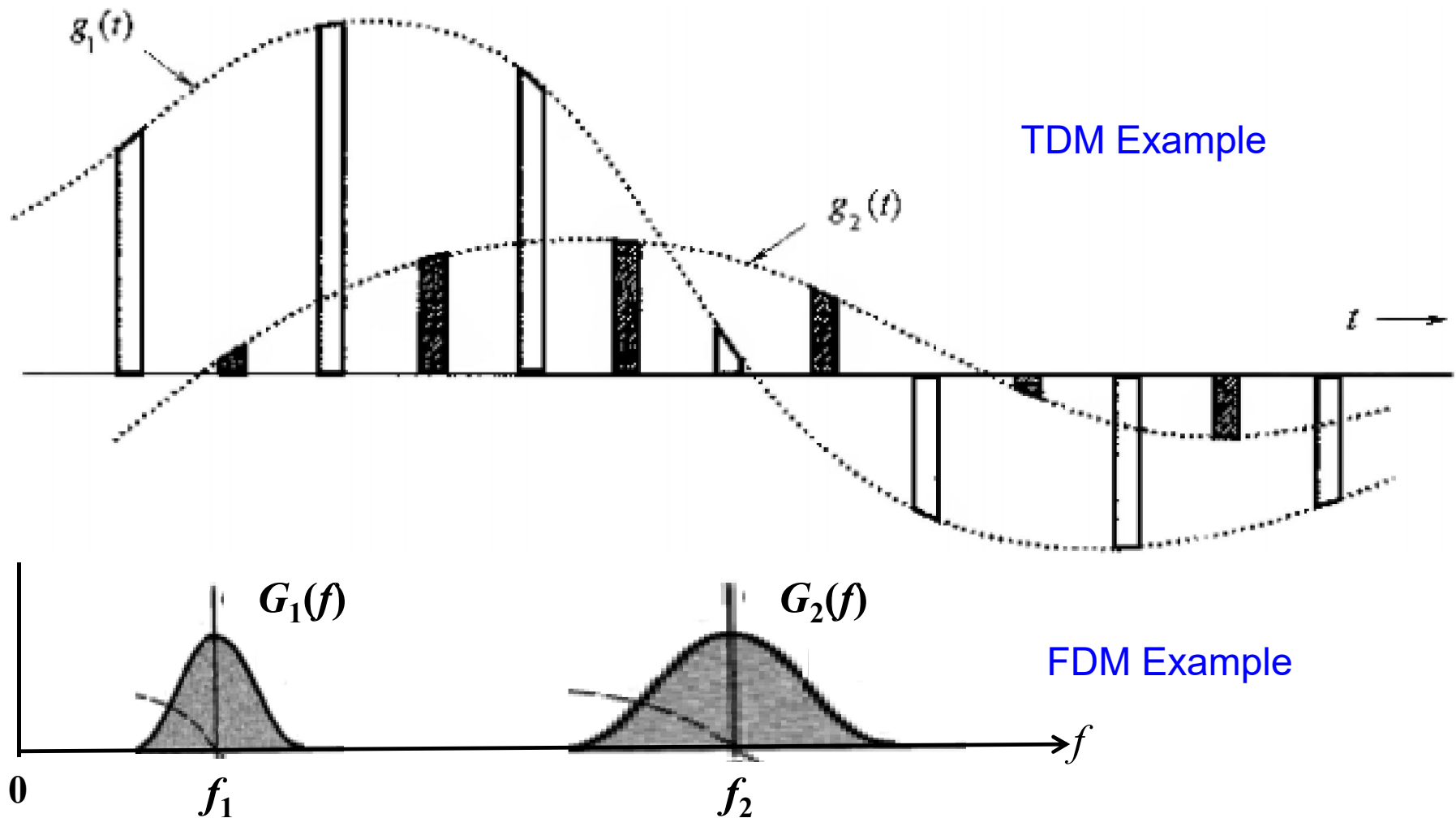
# Application of Sampling Theorem

TDM: Pulses from multiple signals are interweaved on the same channel

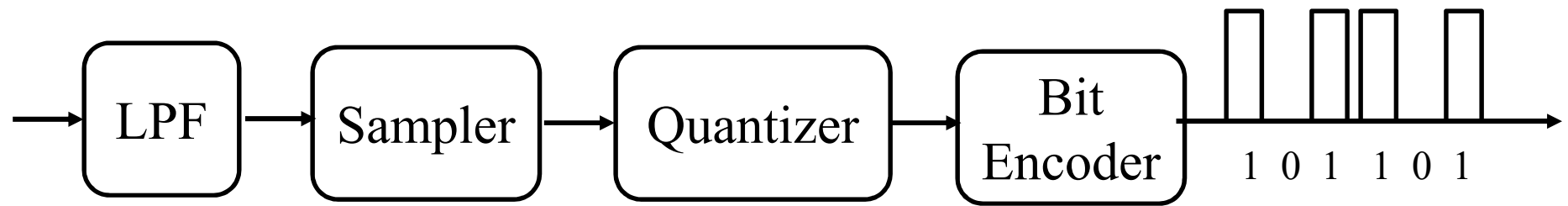


# Application of Sampling Theorem

TDM: dual of FDM where different signals share channel bandwidth



# Pulse Code Modulation (PCM)



PCM system: basically an ADC

Two major Steps:

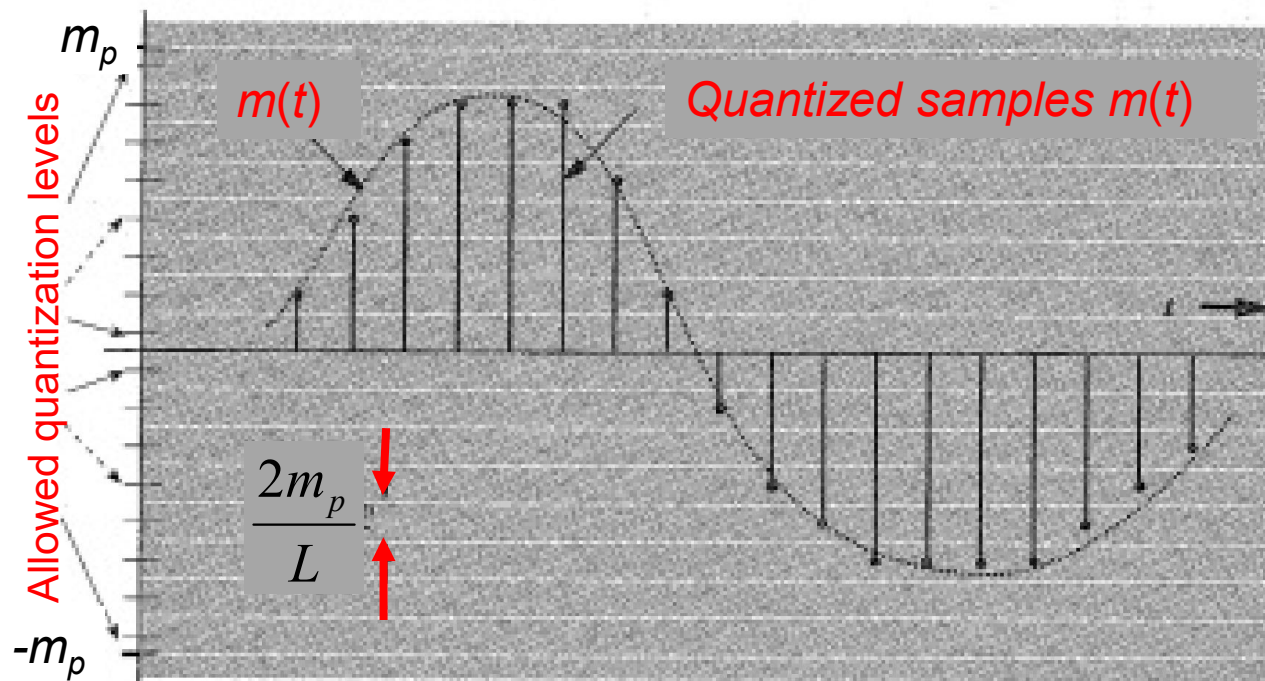
- Sampling and
- quantizing

# Analog to Digital Conversion of Message Signal

- 2 major steps
  - Sampling
  - Quantizing

The range  $(-m_p, m_p)$  is divided into  $L$  sub-intervals, each of magnitude  $\Delta v$

$$\Delta v = \frac{2m_p}{L}$$





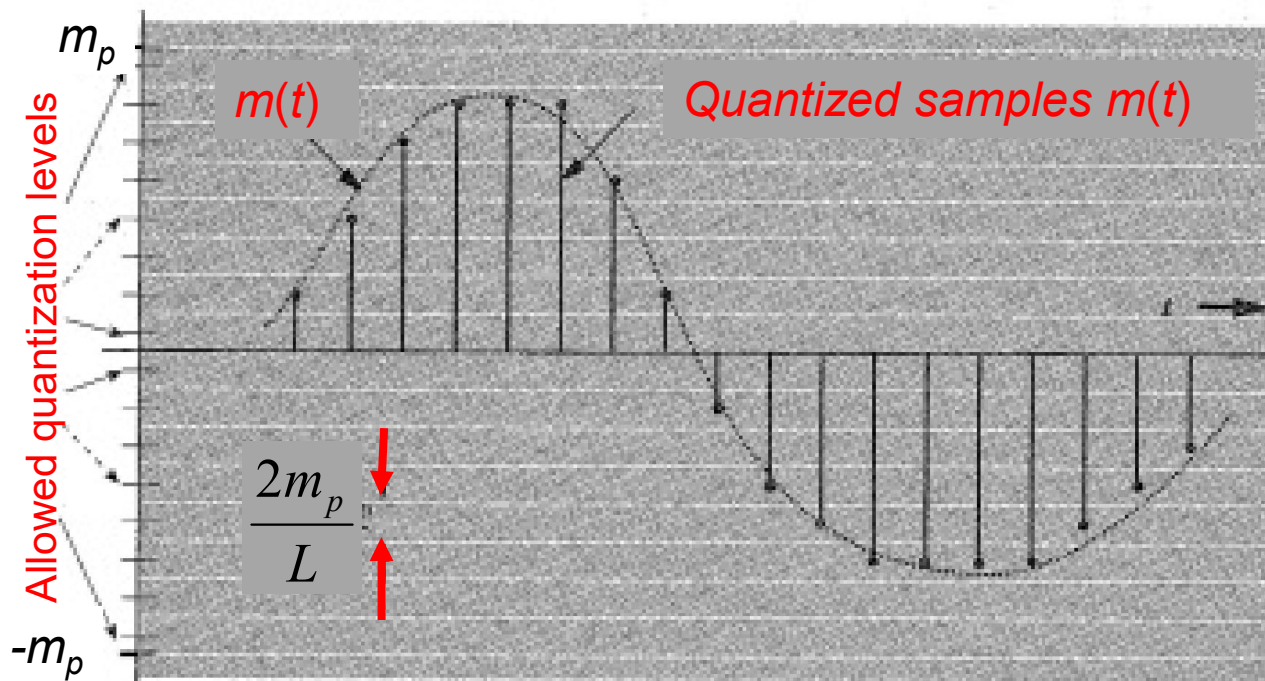
# Analog to Digital Conversion of Message Signal

- 2 major steps
  - Sampling
  - Quantizing

The range  $(-m_p, m_p)$  is divided into  $L$  sub-intervals, each of magnitude  $\Delta v$

$$\Delta v = \frac{2m_p}{L}$$

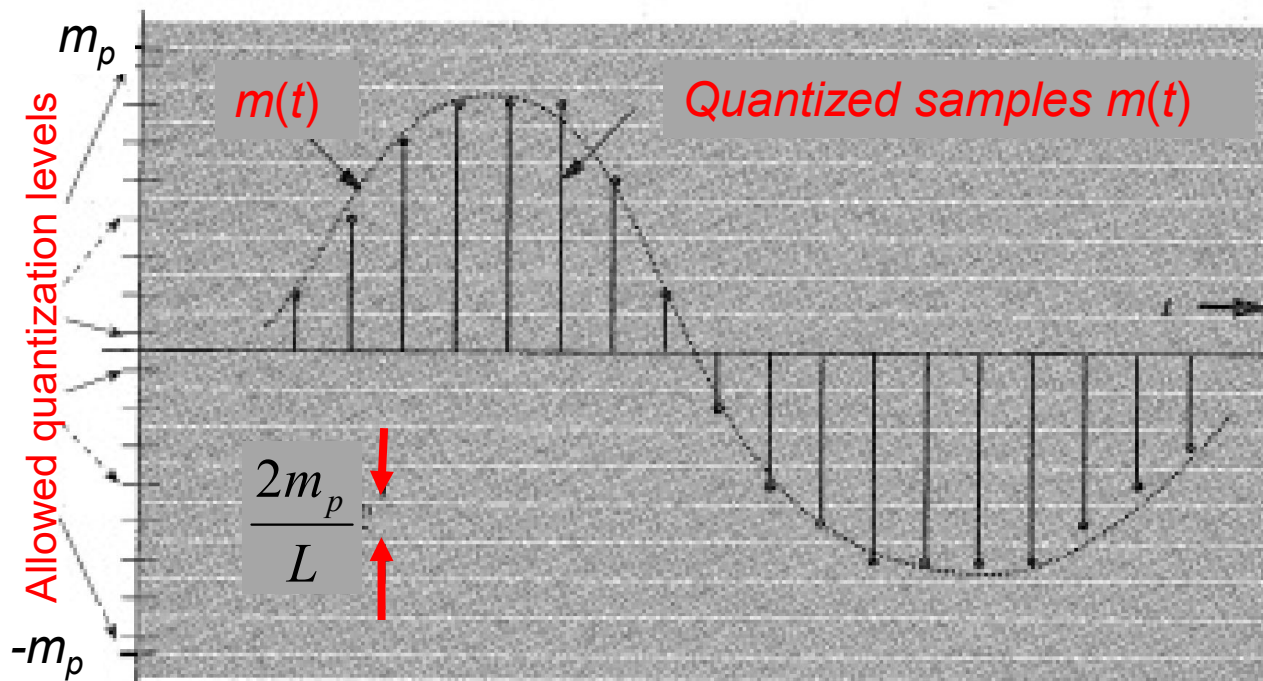
$L$  is known as quantization level



# Analog to Digital Conversion of Message Signal

- 2 major steps
  - Sampling
  - Quantizing

A **sampled value** is placed into one of these  $L$  sub-intervals, thus gets ONE of the  $L$  values

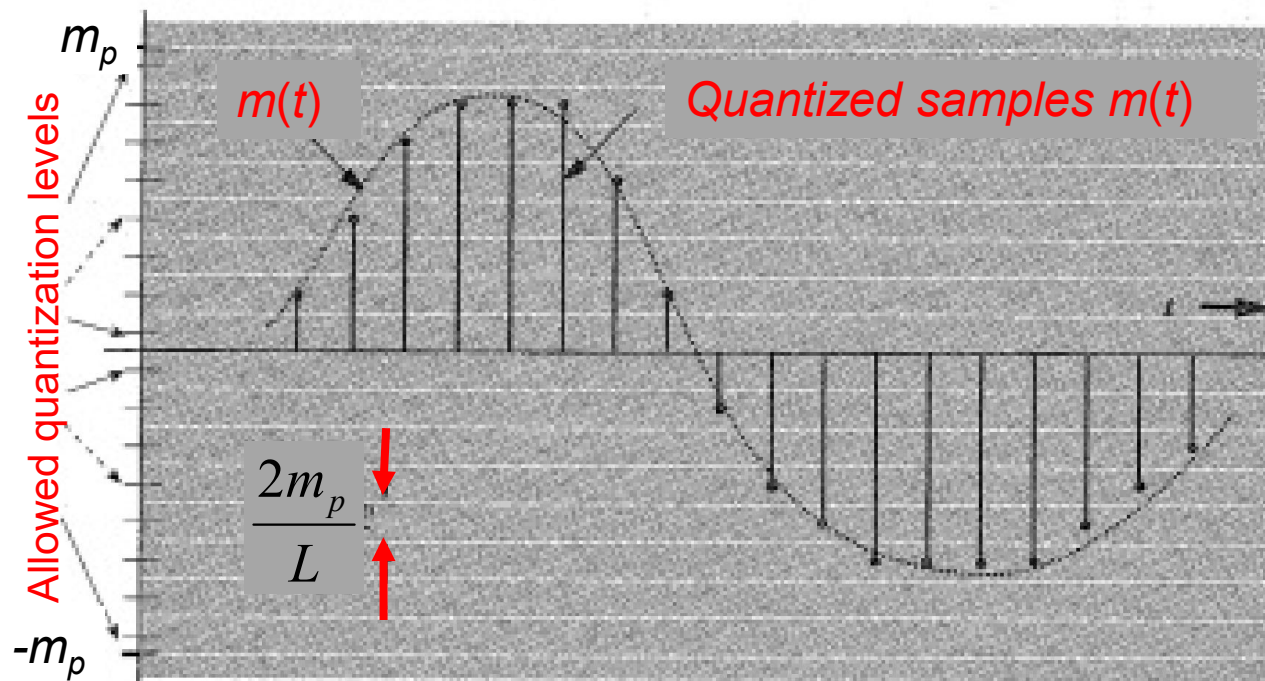


# Analog to Digital Conversion of Message Signal

- 2 major steps
  - Sampling
  - Quantizing

A **sampled value** is placed into one of these  $L$  sub-intervals, thus gets ONE of the  $L$  values

Signal is known as  $L$ -ary digital signal



# Analog to Digital Conversion of Message Signal

$L$ -ary digital signal is converted to binary digital signal using pulse coding

Each of  $L$  values is encoded as a group of binary digits

| Digit | Binary equivalent |
|-------|-------------------|
| 0     | 0000              |
| 1     | 0001              |
| 2     | 0010              |
| 3     | 0011              |
| 4     | 0100              |
| 5     | 0101              |
| 6     | 0110              |
| 7     | 0111              |
| 8     | 1000              |
| 9     | 1001              |
| 10    | 1010              |
| 11    | 1011              |
| 12    | 1100              |
| 13    | 1101              |
| 14    | 1110              |
| 15    | 1111              |

# Analog to Digital Conversion of Message Signal

$L$ -ary digital signal is converted to binary digital signal using pulse coding

Each of  $L$  values is encoded as a group of binary digits

Each bit is transmitted using a distinct pulse shape

| Digit | Binary equivalent | Pulse code waveform |
|-------|-------------------|---------------------|
| 0     | 0000              |                     |
| 1     | 0001              |                     |
| 2     | 0010              |                     |
| 3     | 0011              |                     |
| 4     | 0100              |                     |
| 5     | 0101              |                     |
| 6     | 0110              |                     |
| 7     | 0111              |                     |
| 8     | 1000              |                     |
| 9     | 1001              |                     |
| 10    | 1010              |                     |
| 11    | 1011              |                     |
| 12    | 1100              |                     |
| 13    | 1101              |                     |
| 14    | 1110              |                     |
| 15    | 1111              |                     |

# Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility).

Fidelity is compromised!

# Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

$$B = 3400 \text{ Hz}$$

$$f_s = 8000 > 2B$$

# Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

$$B = 3400 \text{ Hz}$$

$$f_s = 8000 > 2B$$

Quantization level,  $L = 256$  (8 bits)



# Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

$$B = 3400 \text{ Hz}$$

$$f_s = 8000 > 2B$$

Quantization level,  $L = 256$  (8 bits)

$$\text{Data rate} = 8000 * 8 = 64000 \text{ pulse/second} = 64 \text{ Kbps}$$

# Analog to Digital Conversion of Message Signal

Example 2: data rate for compact disc

Fidelity is required!

Audio signal b/w = 20 KHz

$$B = 20000 \text{ Hz}$$

$$f_s = 441000 \text{ Hz} > 2B$$

Quantization level,  $L = 65,536$  (16 bits)

$$\text{Data rate} = 44100 * 16 = 1.4 \text{ Mbps}$$

# **Advantages of Digital Communication**

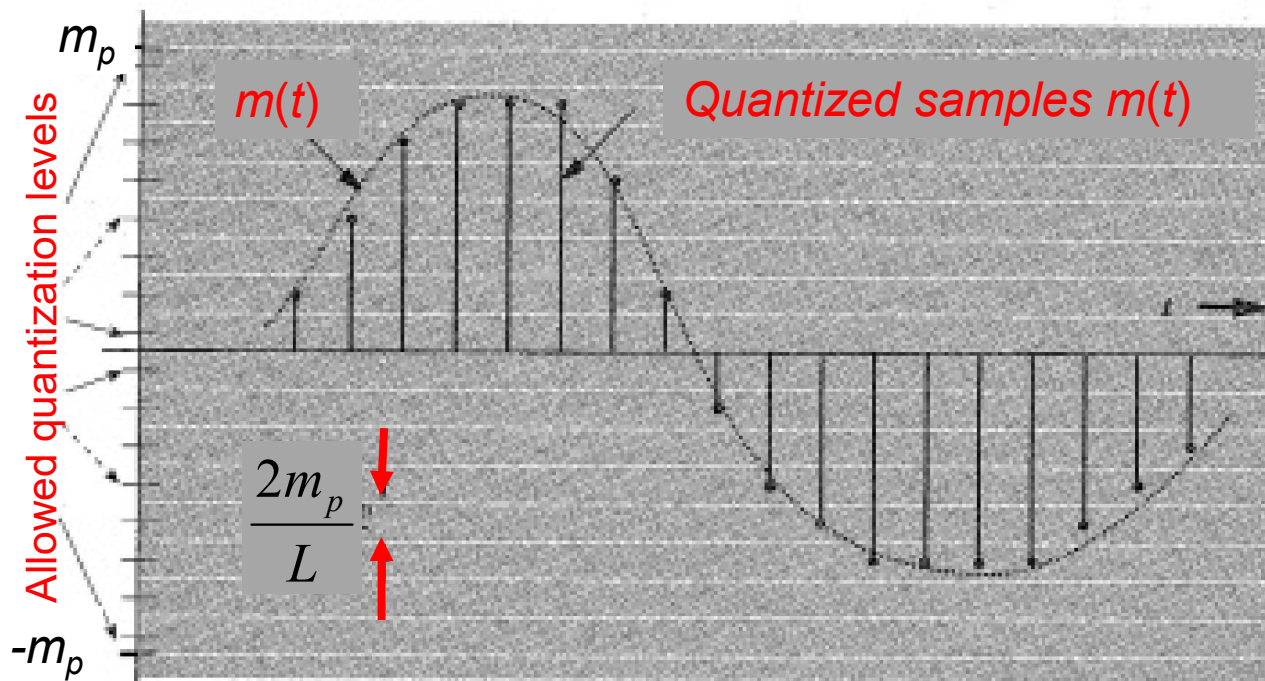
Self Study

# Quantization

The range  $(-m_p, m_p)$  is divided into  $L$  sub-intervals, each of magnitude  $\Delta v$

$$\Delta v = \frac{2m_p}{L}$$

$L$  is known as quantization level



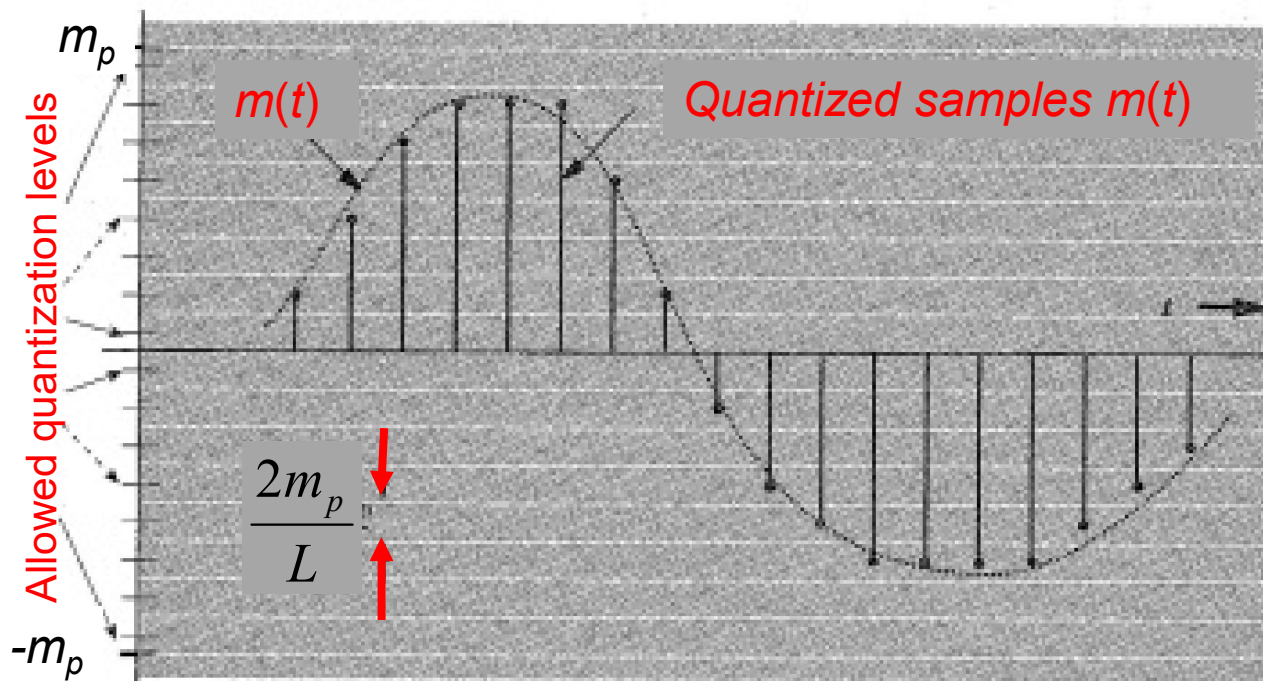
# Quantization

$m_p$  is NOT the signal PEAK, rather is it's the LIMIT of the quantizer

The range  $(-m_p, m_p)$  is divided into  $L$  sub-intervals, each of magnitude  $\Delta v$

$$\Delta v = \frac{2m_p}{L}$$

$L$  is known as quantization level



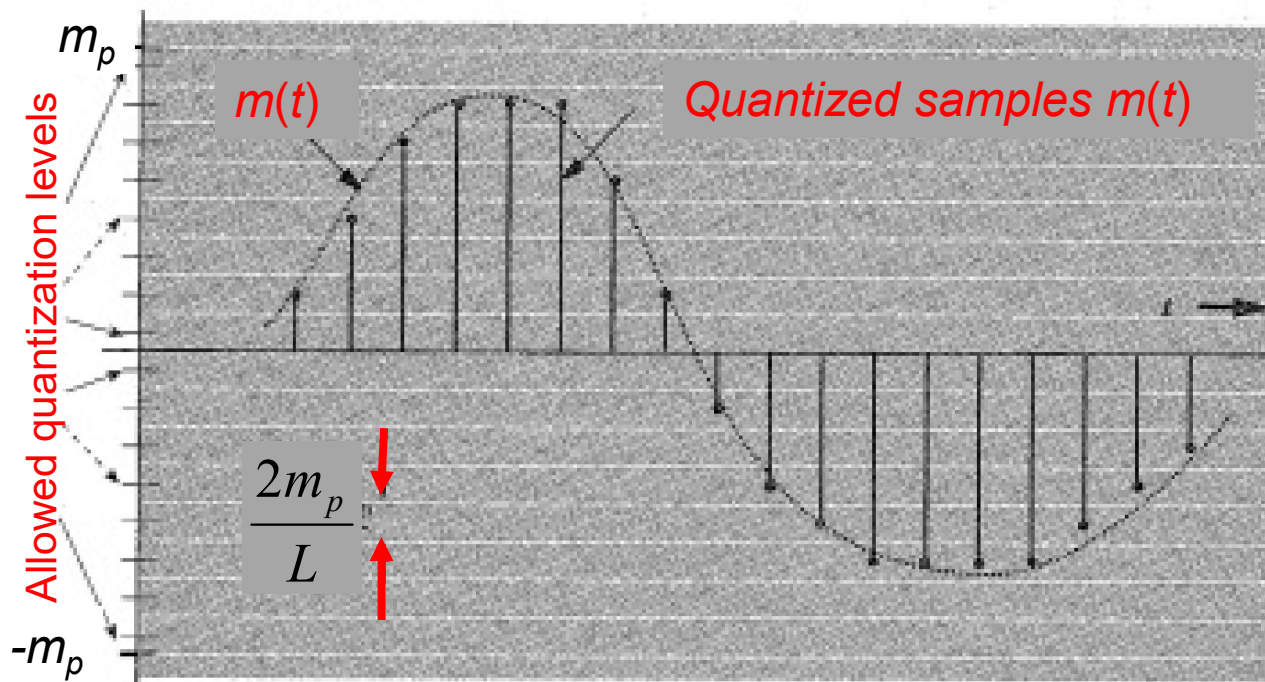
# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

The range  $(-m_p, m_p)$  is divided into  $L$  sub-intervals, each of magnitude  $\Delta v$

$$\Delta v = \frac{2m_p}{L}$$

$L$  is known as quantization level



# Quantization

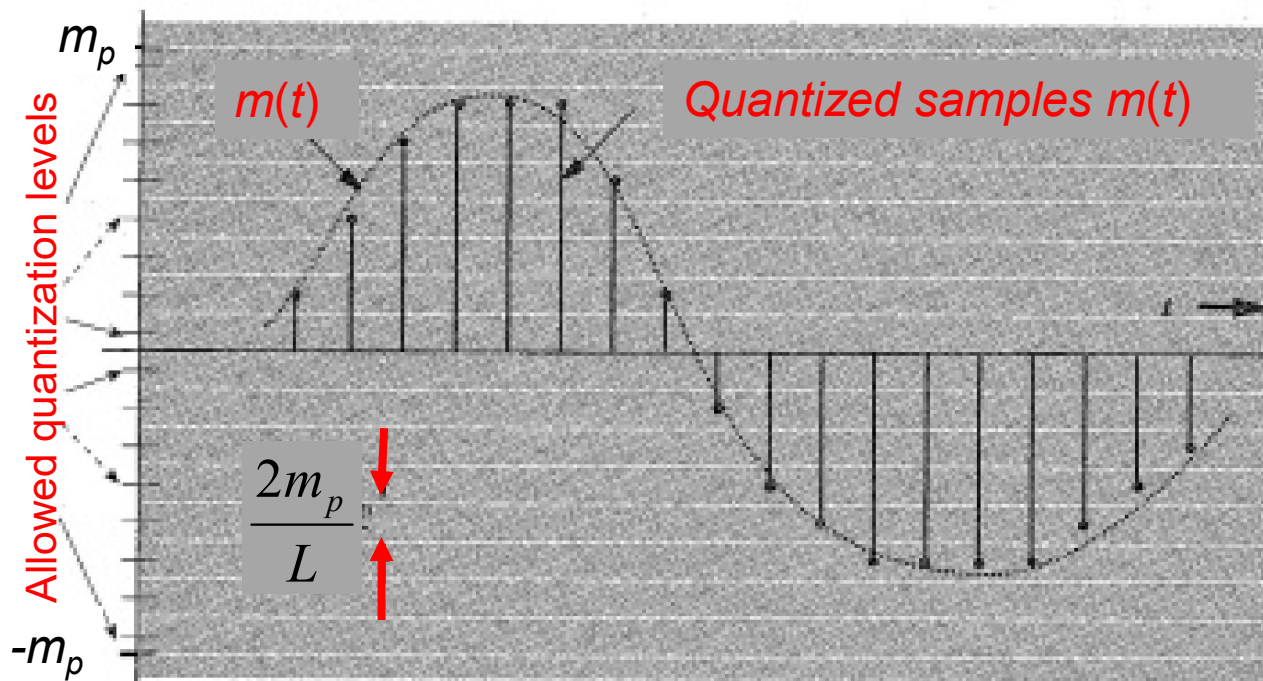
$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

The range  $(-m_p, m_p)$  is divided into  $L$  sub-intervals, each of magnitude  $\Delta v$

$$\Delta v = \frac{2m_p}{L}$$

$L$  is known as quantization level



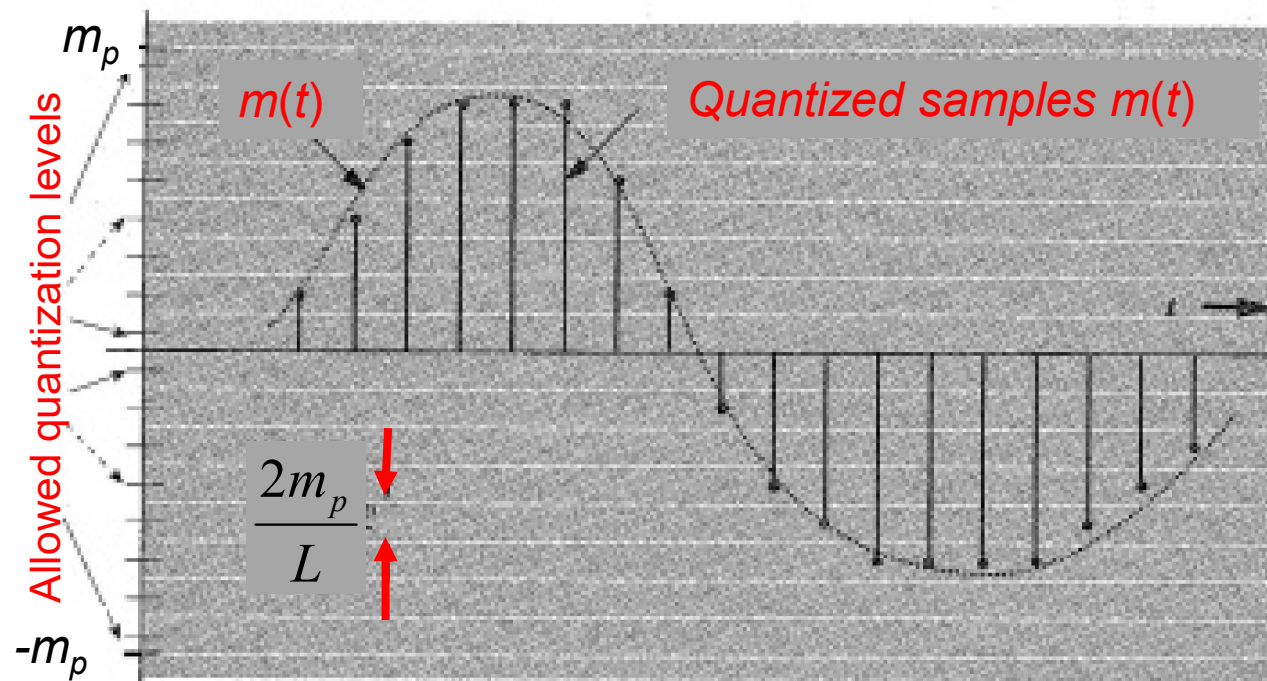


# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in  $(-\Delta v/2, \Delta v/2)$



$$\Delta v = \frac{2m_p}{L}$$

$L$  is known as quantization level

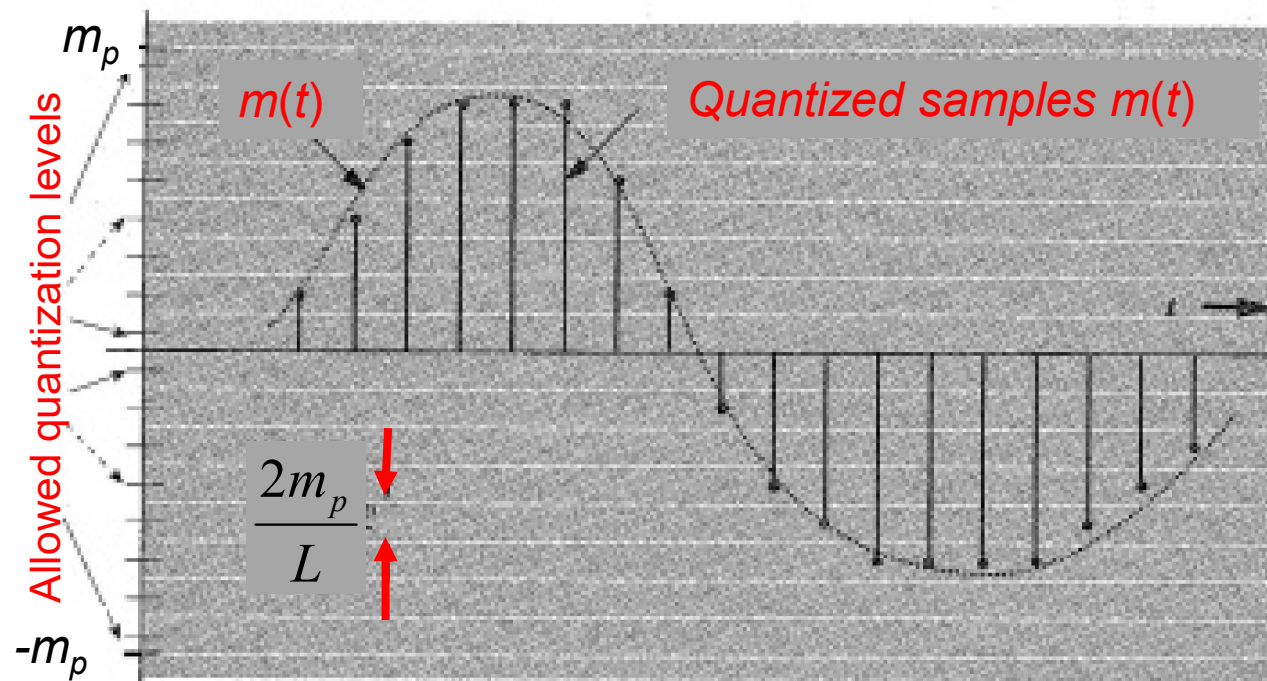


# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in  $(-\Delta v/2, \Delta v/2)$



$$\begin{aligned} &\hat{m}(kT_s) + \Delta v / 2 \\ &\hat{m}(kT_s) \\ &\hat{m}(kT_s) - \Delta v / 2 \end{aligned}$$

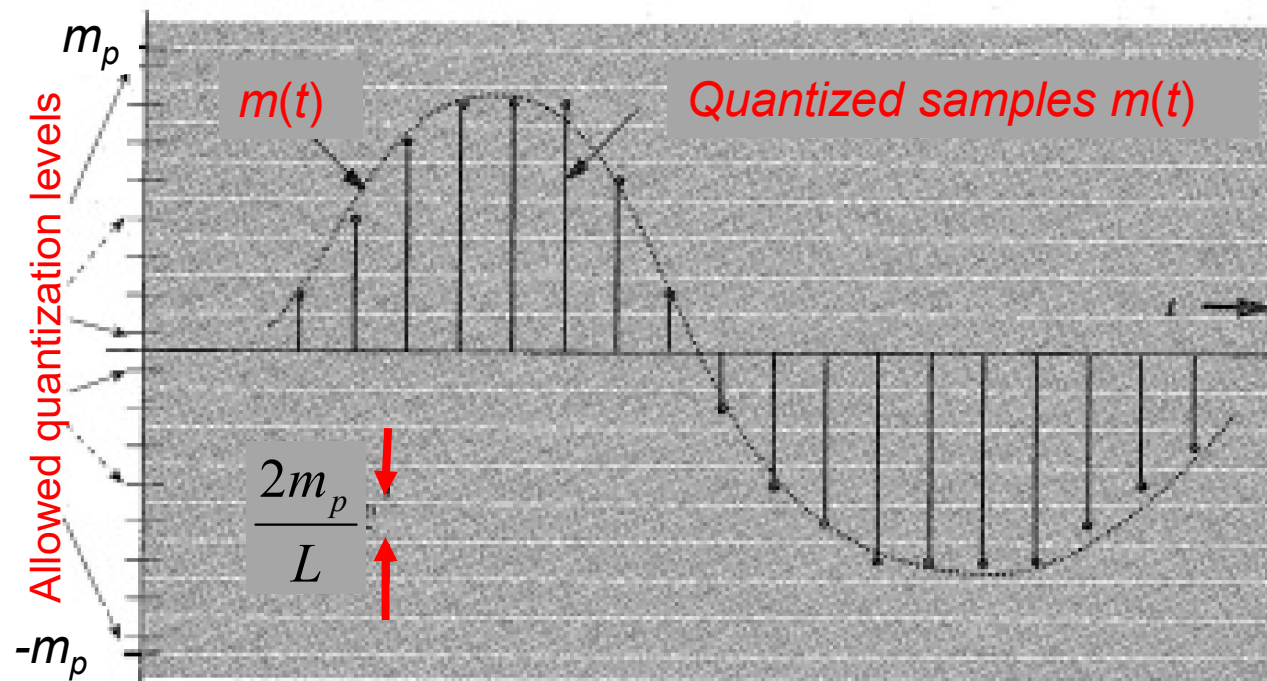
$$\Delta v = \frac{2m_p}{L}$$

# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in  $(-\Delta v/2, \Delta v/2)$



$$\begin{array}{c} \hat{m}(kT_s) + \Delta v / 2 \\ \hline \hat{m}(kT_s) \\ \hline \hat{m}(kT_s) - \Delta v / 2 \end{array}$$

$m(kT_s)$

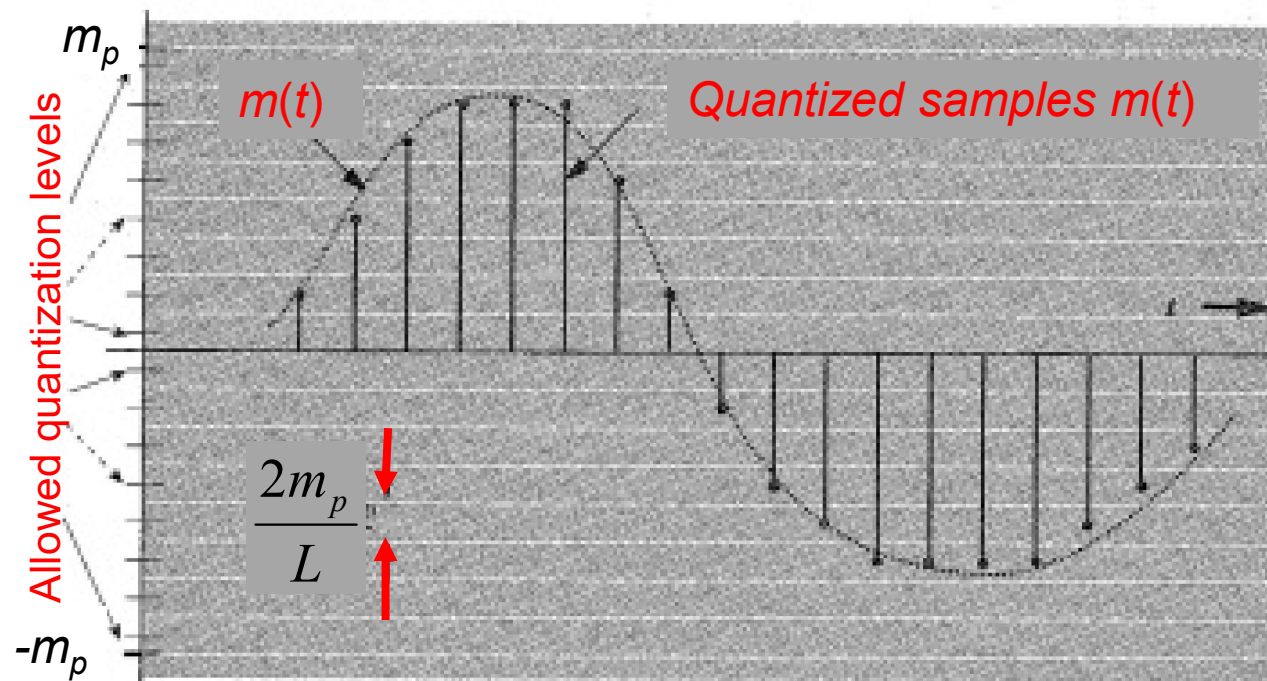
$$\Delta v = \frac{2m_p}{L}$$

# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in  $(-\Delta v/2, \Delta v/2)$



$$\begin{aligned} &\hat{m}(kT_s) + \Delta v / 2 \\ &\hat{m}(kT_s) \\ &\hat{m}(kT_s) - \Delta v / 2 \end{aligned}$$

$m(kT_s)$

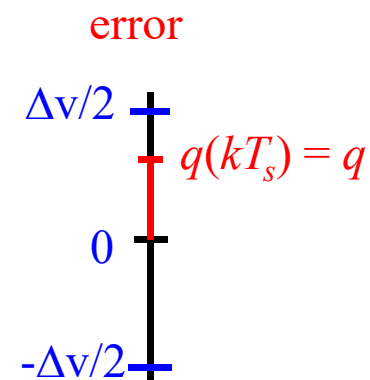
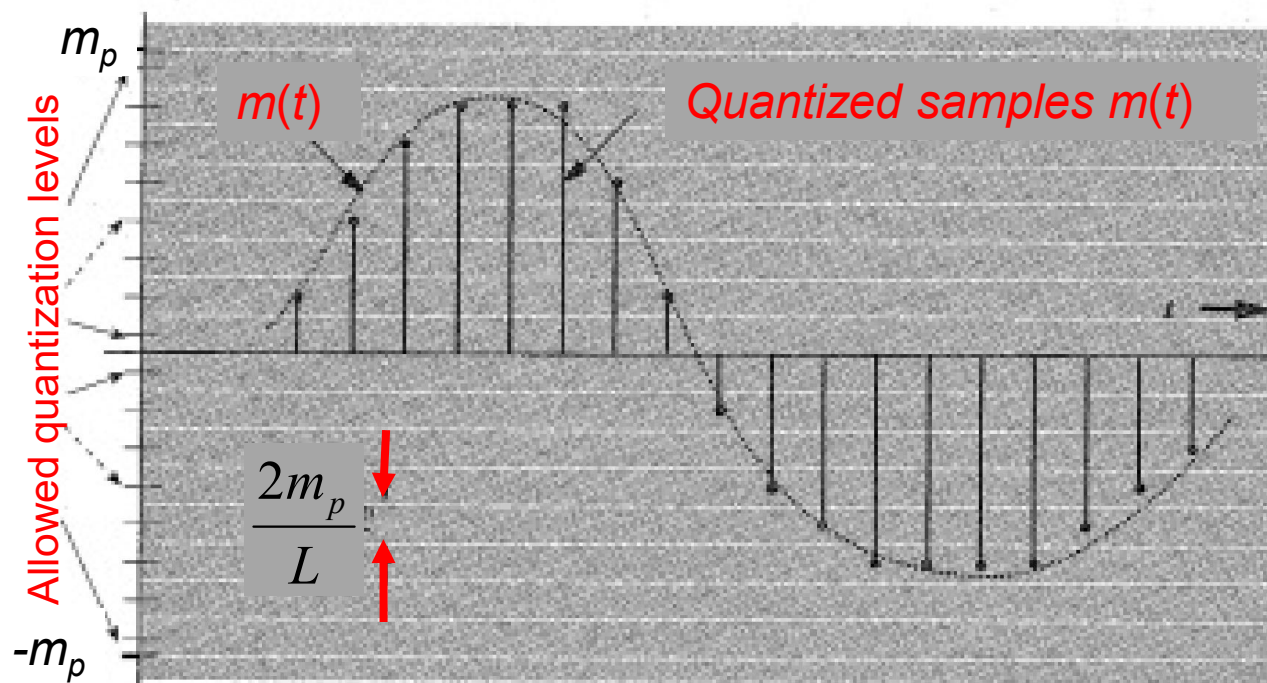
$$\Delta v = \frac{2m_p}{L}$$

# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in  $(-\Delta v/2, \Delta v/2)$



$$\Delta v = \frac{2m_p}{L}$$

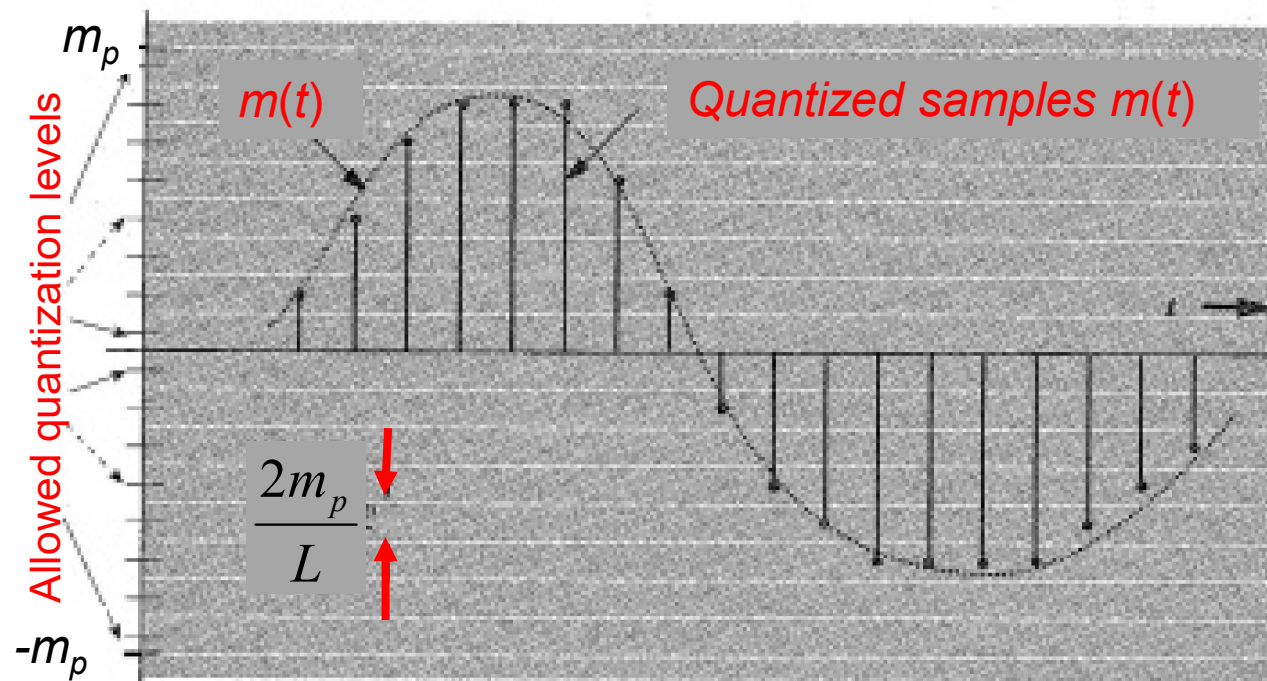


# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in  $(-\Delta v/2, \Delta v/2)$



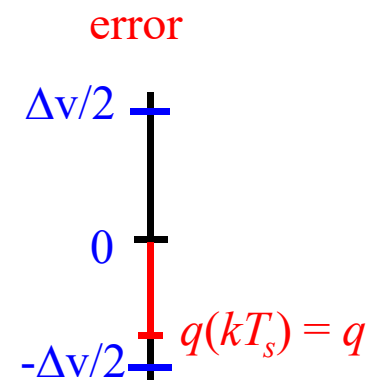
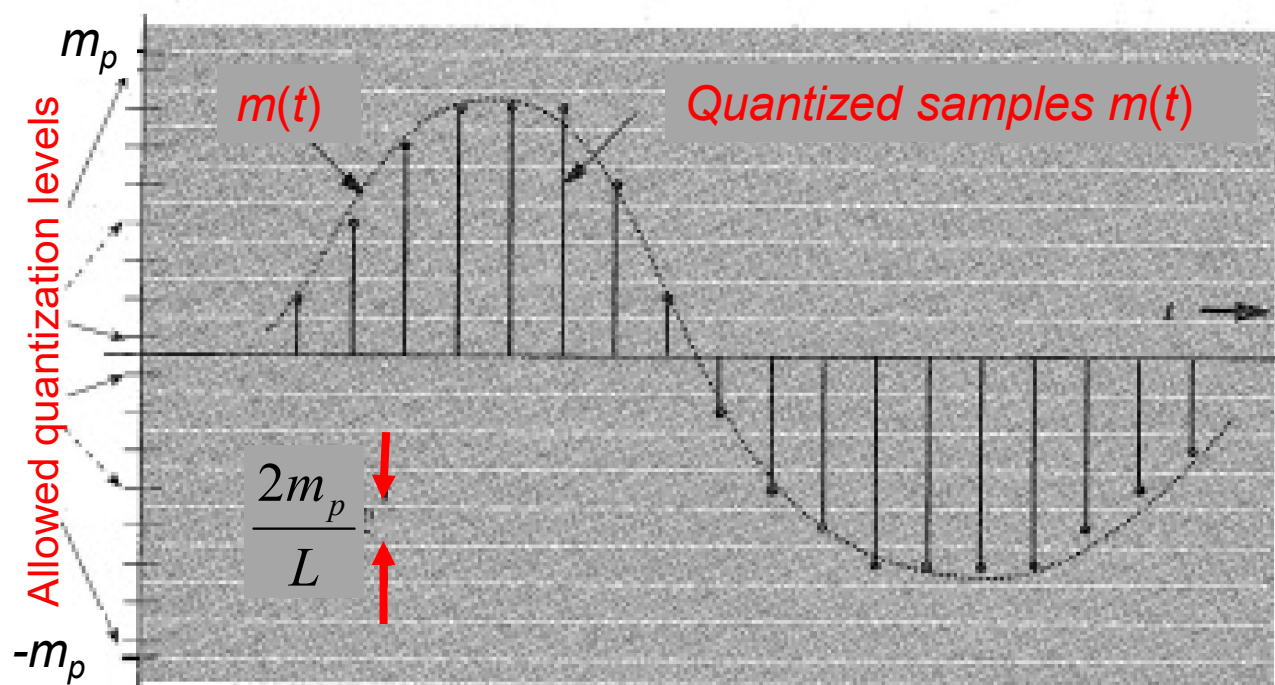
$$\Delta v = \frac{2m_p}{L}$$

# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

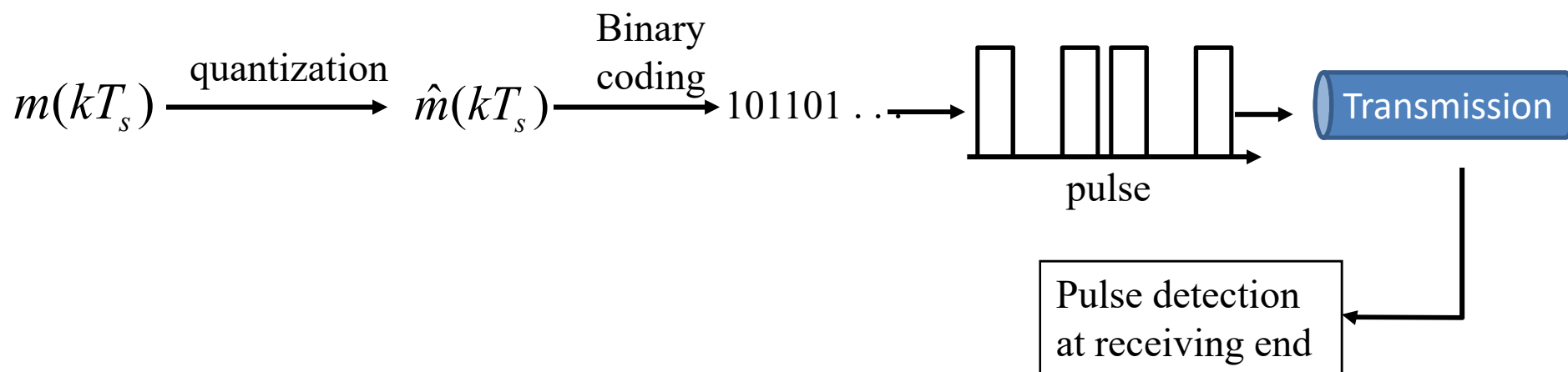
quantization error is unavoidable which lies in  $(-\Delta v/2, \Delta v/2)$



$$\Delta v = \frac{2m_p}{L}$$

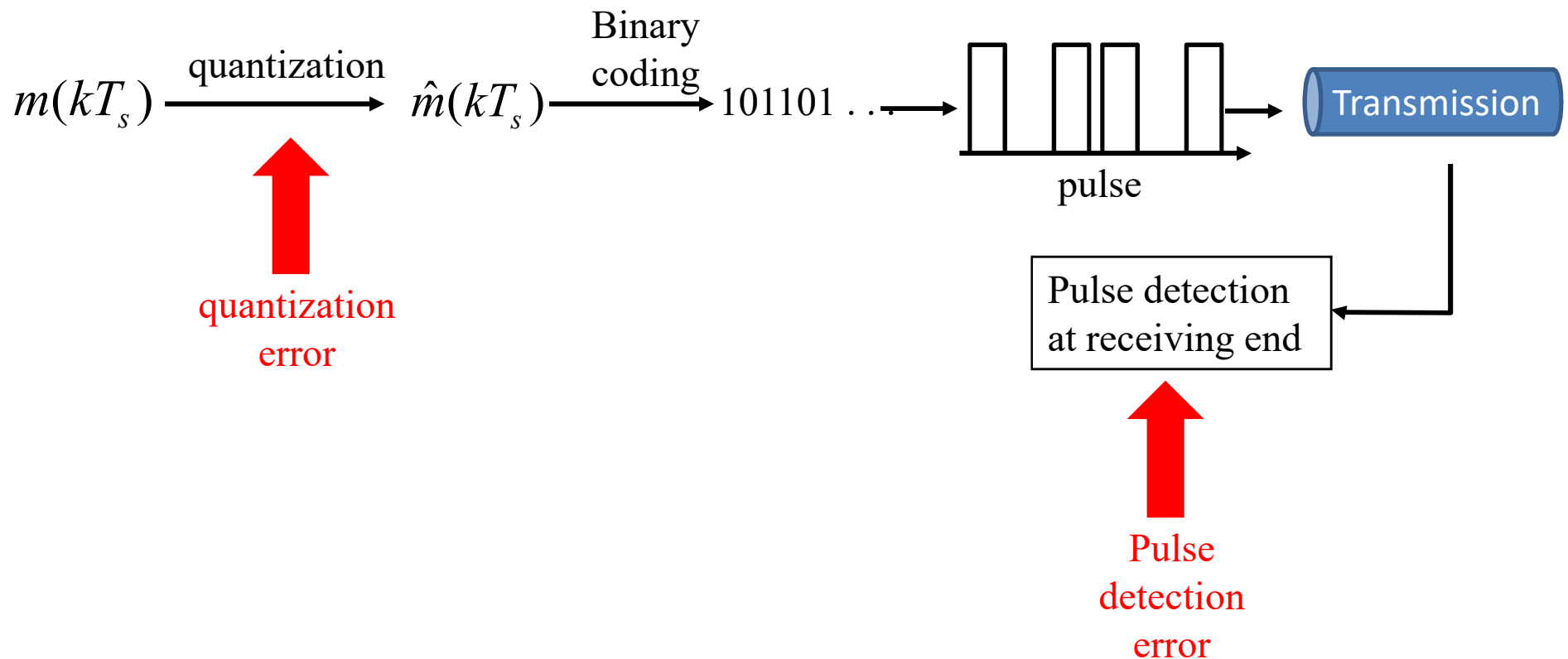
# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies



# Quantization

$k$ -th sample value  $m(kT_s)$  is replaced by the midpoint of an interval where it lies





# Quantization

If there were no quantization error,

$$m(t) = \sum_k m(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

# Quantization

If there were no quantization error,

$$m(t) = \sum_k m(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Due to quantization error,

$$\hat{m}(t) = \sum_k \hat{m}(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

# Quantization

If there were no quantization error,

$$m(t) = \sum_k m(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Due to quantization error,

$$\hat{m}(t) = \sum_k \hat{m}(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Quantization error  $q(t)$ ,

$$q(t) = \hat{m}(t) - m(t)$$

# Quantization

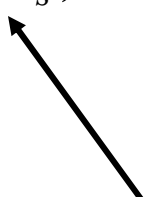
Quantization error or quantization noise or undesired signal,

$$\begin{aligned} q(t) &= \sum_k [\hat{m}(kT_s) - m(kT_s)] \text{sinc}(2\pi Bt - k\pi) \\ &= \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \end{aligned}$$

# Quantization

Quantization noise,

$$\begin{aligned} q(t) &= \sum_k [\hat{m}(kT_s) - m(kT_s)] \text{sinc}(2\pi Bt - k\pi) \\ &= \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \end{aligned}$$



$q(kT_s)$  = Quantization  
error for  $k$ th sample

# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned} \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \end{aligned}$$

# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [a_1 + a_2 + a_3 + \dots]^2 dt\end{aligned}$$

# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [a_1 + a_2 + a_3 + \dots]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [a_1^2 + a_2^2 + a_3^2 + \dots + 2a_1a_2 + 2a_1a_3 + 2a_1a_4 + \dots] dt\end{aligned}$$



# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ a_1^2 + a_2^2 + a_3^2 + \cdots + 2a_1a_2 + 2a_1a_3 + 2a_1a_4 + \cdots \right] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k a_k^2 + 2 \sum_{m \neq n} a_m a_n \right] dt\end{aligned}$$

# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k a_k^2 + 2 \sum_{m \neq n} a_m a_n \right] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k q^2(kT_s) \text{sinc}^2(2\pi Bt - k\pi) \right] dt \\ &\quad + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ 2 \sum_{m \neq n} q(mT_s) q(nT_s) \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) \right] dt\end{aligned}$$

# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}
 \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ \sum_k q^2(kT_s) \text{sinc}^2(2\pi Bt - k\pi) \right] dt \\
 &+ \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[ 2 \sum_{m \neq n} q(mT_s) q(nT_s) \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) \right] dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_k q^2(kT_s) \int_{-T/2}^{T/2} \text{sinc}^2(2\pi Bt - k\pi) dt \\
 &+ \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{m \neq n} q(mT_s) q(nT_s) \int_{-T/2}^{T/2} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt
 \end{aligned}$$

# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned} \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_k q^2(kT_s) \int_{-T/2}^{T/2} \text{sinc}^2(2\pi Bt - k\pi) dt \\ &+ \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{m \neq n} q(mT_s) q(nT_s) \int_{-T/2}^{T/2} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt \end{aligned}$$

# Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned} \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_k q^2(kT_s) \int_{-T/2}^{T/2} \text{sinc}^2(2\pi Bt - k\pi) dt \\ &+ \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{m \neq n} q(mT_s) q(nT_s) \int_{-T/2}^{T/2} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt \end{aligned}$$

We can prove that,

$$\int_{-\infty}^{\infty} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt = \begin{cases} 0 & m \neq n \\ \frac{1}{2B} & m = n \end{cases}$$

# Quantization

Power or Mean square of Quantization noise,

$$\overline{q^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{2BT} \sum_k q^2(kT_s)$$

# Quantization

Power or Mean square of Quantization noise,

$$\overline{q^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{2BT} \sum_k q^2(kT_s)$$

As sampling frequency  $f_s = 2B$ ,

$2BT = \text{total no. of samples over averaging time } T$

RHS is the mean of the square of quantization error

# Quantization

Power or Mean square of Quantization noise,

$$\overline{q^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{2BT} \sum_k q^2(kT_s)$$

As sampling frequency  $f_s = 2B$ ,

$2BT = \text{total no. of samples over averaging time } T$

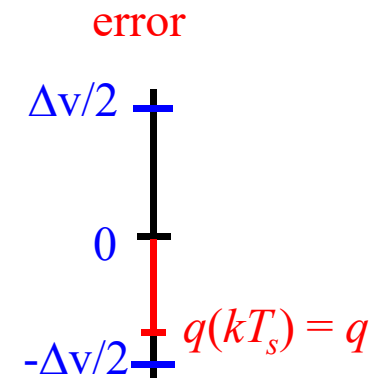
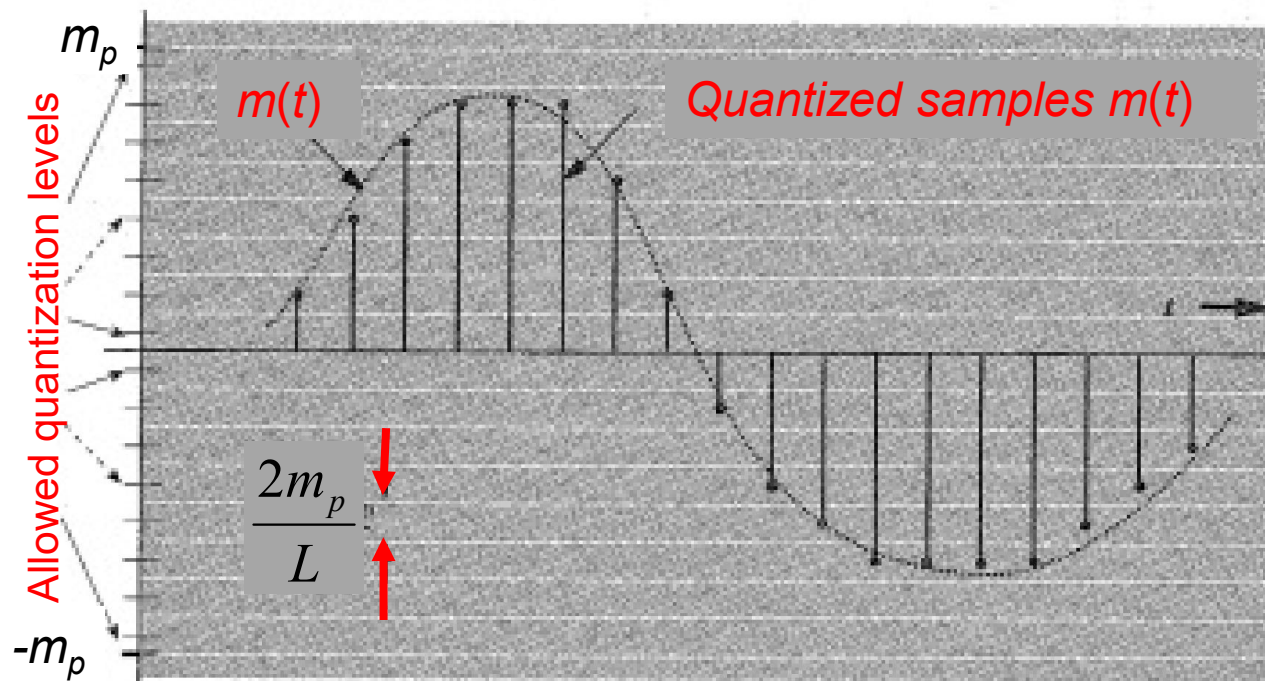
RHS is the mean of the square of quantization error

Therefore, power of quantization noise = mean square quantization error



# Quantization

We know, quantization error  $q$  lies in  $(-\Delta v/2, \Delta v/2)$

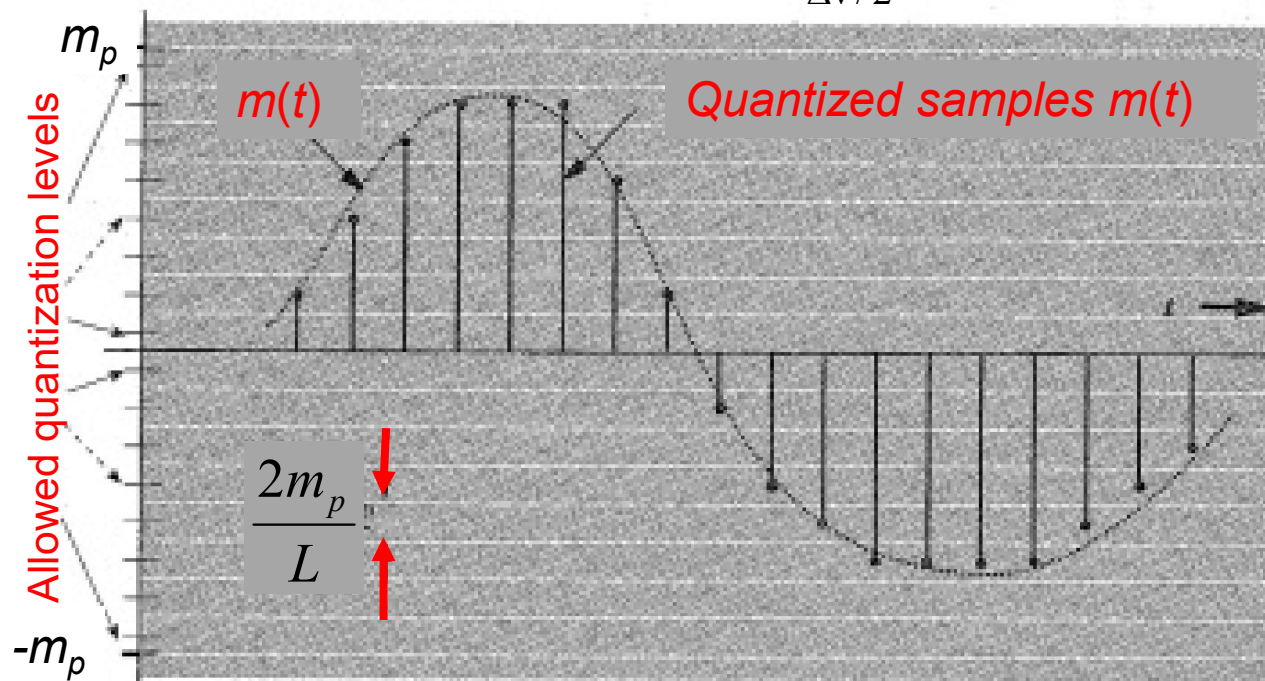


$$\Delta v = \frac{2m_p}{L}$$

# Quantization

Mean square quantization error is given by

$$\overline{q^2} = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq$$



# Quantization

Mean square quantization error is given by

$$\tilde{\tilde{q}}^2 = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12}$$

# Quantization

Mean square quantization error is given by

$$\overline{\overline{q^2}} = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12} = \frac{m_p^2}{3L^2}$$

$$\text{where, } \Delta v = \frac{2m_p}{L}$$

# Quantization

Mean square quantization error is given by

$$\tilde{\tilde{q}}^2 = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12} = \frac{m_p^2}{3L^2}$$

We proved, power of quantization noise ( $N_0$ ) = mean square quantization error

# Quantization

Mean square quantization error is given by

$$\overline{q^2} = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12} = \frac{m_p^2}{3L^2}$$

We proved, power of quantization noise ( $N_0$ ) = mean square quantization error

$$N_0 = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$$

# Quantization

power of quantization noise ( $N_0$ ) = mean square quantization error

$$N_0 = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$$

Assume, power of message signal ( $S_0$ ) is given by  $S_0 = \overline{m^2(t)}$

# Quantization

power of quantization noise ( $N_0$ ) = mean square quantization error

$$N_0 = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$$

Assume, power of message signal ( $S_0$ ) is given by  $S_0 = \overline{m^2(t)}$

Signal-to-noise ratio (SNR) is

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$



# Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- Higher  $SNR$  means higher quality of received signal
- $L$  increases  $SNR$
- Higher limit of quantizer ( $m_p$ ) decreases  $SNR$

# Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- Higher  $SNR$  means higher quality of received signal
- $L$  increases  $SNR$
- Higher limit of quantizer ( $m_p$ ) decreases  $SNR$
- $SNR$  is linear function of signal power,  $S_0 = \overline{m^2(t)}$

# Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- $SNR$  is linear function of signal power,  $S_0 = \overline{m^2(t)}$

$S_0$  varies

- from speaker to speaker
- due to different length of connecting circuits

# Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- $SNR$  is linear function of signal power,  $S_0 = \overline{m^2(t)}$

$S_0$  varies

- from speaker to speaker
- due to different length of connecting circuits

For these reasons,

- $SNR$  varies widely
- Quality of received signal deteriorates remarkably for soft speakers

# Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- $SNR$  is linear function of signal power,  $S_0 = \overline{m^2(t)}$

$S_0$  varies

- from speaker to speaker
- due to different length of connecting circuits

For these reasons,

- $SNR$  varies widely
- Quality of received signal deteriorates remarkably for soft speakers

However statistically,

- Small amplitudes (soft speakers) predominate in speech
- Larger amplitudes (loud speakers) are less frequent

# Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

## Causes of the problem

- All amplitudes are **uniformly** quantized with **equal step size**  $\Delta v = \frac{2m_p}{L}$
- **Variation in amplitudes is NOT considered**

# Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

## Causes of the problem

- All amplitudes are **uniformly** quantized with **equal step size**  $\Delta v = \frac{2m_p}{L}$
- Variation in amplitudes is **NOT** considered

Quantization noise  $N_0 = (\Delta v)^2/12$  is directly **proportional to square of step size**  $\Delta v$

$$N_0 \propto (\Delta v)^2$$